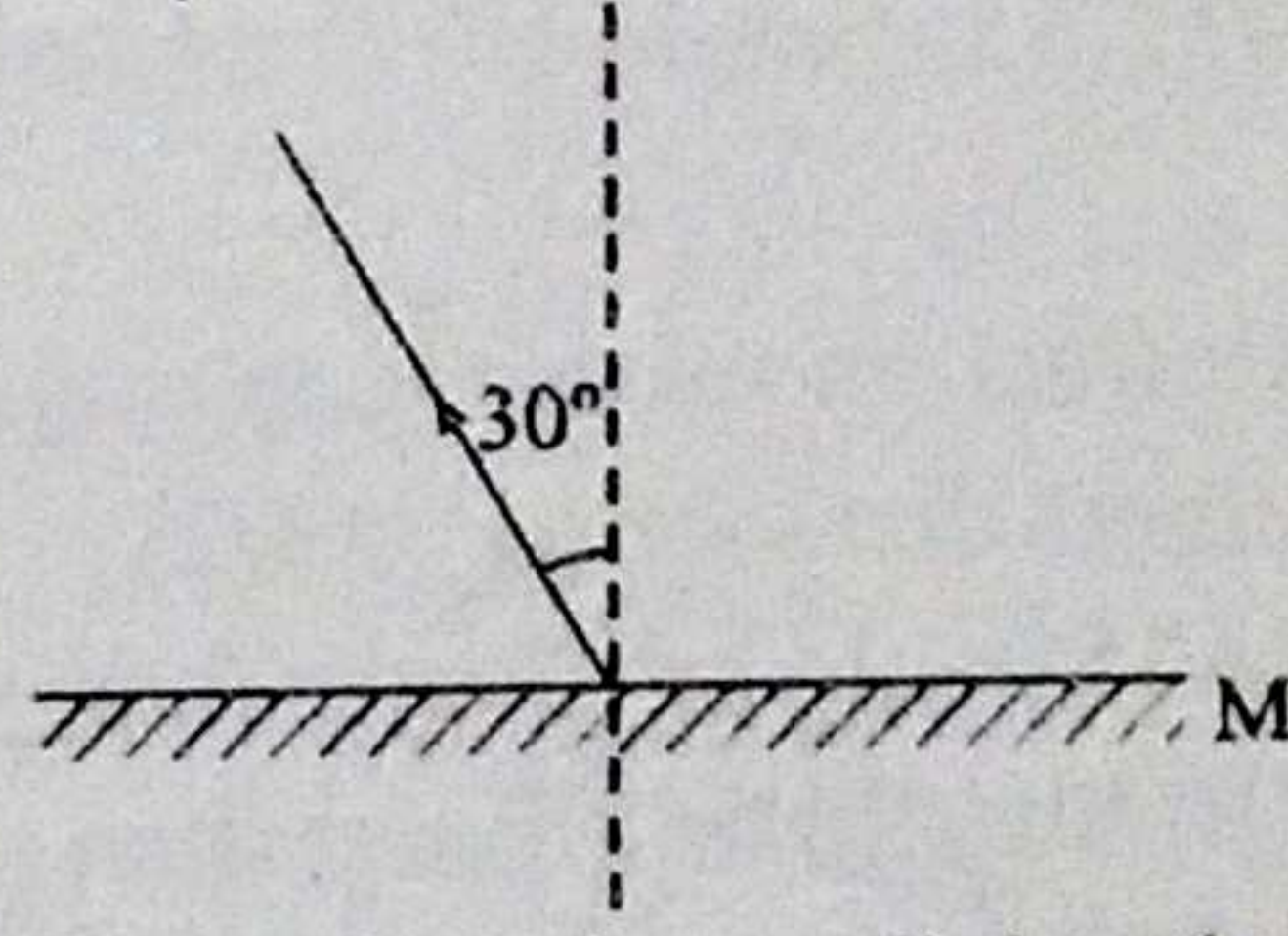
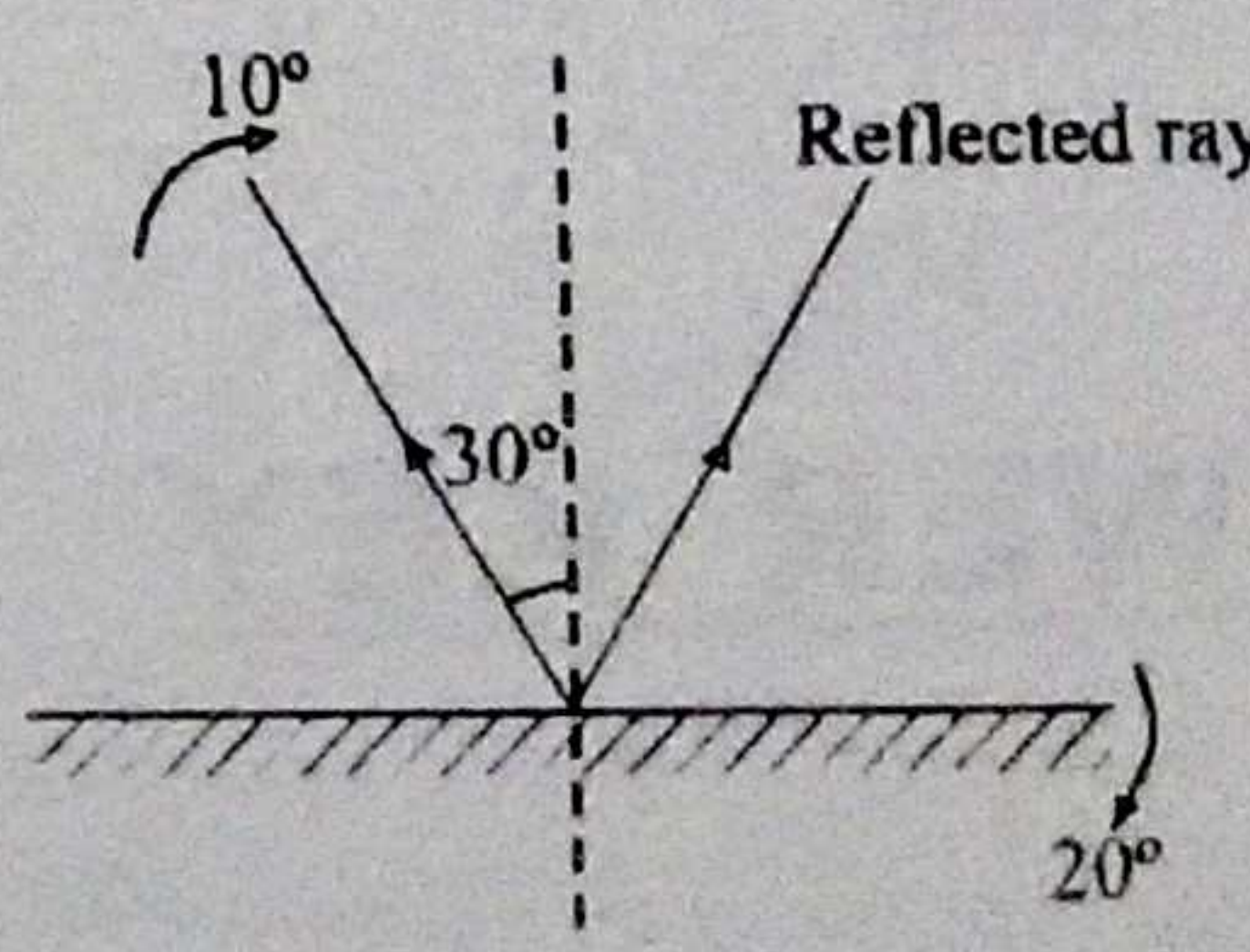


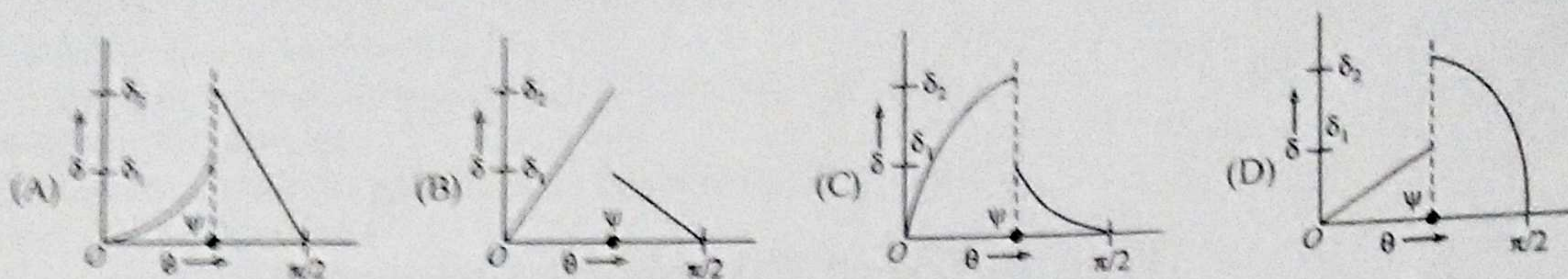
Geometrical Optics

Type of CPP - Objective & Subjective

Level - MODERATE

1. A point object is kept between a plane mirror and a concave mirror facing each other. The distance between the mirrors is 22.5 cm. The radius of curvature of the concave mirror is 20 cm. What should be the distance of the object from the concave mirror so that after two successive reflections the final image is formed on the object itself?
[Consider first reflection from concave mirror]:
(A) 5 cm (B) 15 cm (C) 10 cm (D) 7.5 cm
2. Half the surface of a transparent sphere of refractive index 2 is silvered. A narrow, parallel beam of light is incident on the unsilvered surface, symmetrically with respect to the silvered part. The light finally emerging from the sphere will be a :
(A) Parallel beam (B) Converging beam (C) Slightly divergent beam (D) Widely divergent beam
3. An object AB is kept horizontally. How will you place a plane mirror to get its image vertically. Draw its ray diagram.
4. Find the angle of deviation (both clockwise and anticlockwise) suffered by a ray incident on a plane mirror, at an angle of incidence 30° .

5. Two plane mirrors are inclined to each other at an angle 60° . If a ray of light incident on the first mirror is parallel to the second mirror, it is reflected from the second mirror :
(A) Perpendicular to the first mirror (B) Parallel to the first mirror
(C) Parallel to the second mirror (D) Perpendicular to the second mirror
6. A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that the end farther from the pole is 20 cm away from it. Find the length of the image :
7. A particle is moving towards a fixed spherical mirror. The image :
(A) Must move away from the mirror
(B) Must move towards the mirror
(C) May move towards the mirror
(D) Will move towards the mirror, only if the mirror is convex.
8. Figure shows a plane mirror onto which a light ray is incident light ray is turned by 10° and the mirror by 20° , as shown, find the angle turned by the reflected ray.

9. An object is placed at 20 cm from a convex mirror of focal length 20 cm. The distance of the image from the pole of the mirror is :
(A) infinite (B) 10 cm (C) 15 cm (D) 40 cm
10. Which of the following statements are incorrect for spherical mirrors :
(A) A concave mirror forms only virtual images for any position of real object
(B) A convex mirror forms only virtual images for any position of a real object
(C) A concave mirror forms only a virtual diminished image of an object placed between its pole and the focus
(D) A convex mirror forms a virtual magnified image of an object placed between its pole and the focus

11. A ray of light travels from a medium of refractive index μ to air. Its angle of incidence in the medium is θ , measured from the normal to the boundary, and its angle of deviation is δ . δ is plotted against θ . Which of the following best represents the resulting curve?



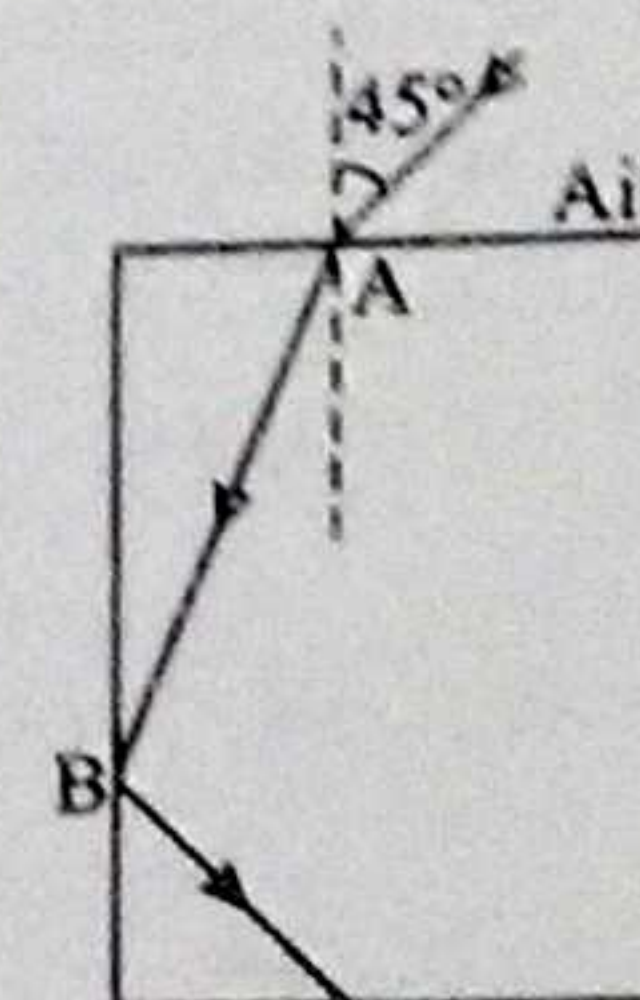
12. In the previous question :

(A) $\psi = \sin^{-1} \left(\frac{1}{\mu} \right)$ (B) $\psi = \frac{\pi}{2} - \sin^{-1} \left(\frac{1}{\mu} \right)$ (C) $\frac{\delta_2}{\delta_1} = \mu$ (D) $\frac{\delta_2}{\delta_1} = 2$

13. For refraction through a small angled prism, the angle of deviation ($n_s < n_p$):

- (A) Increases with the increase in R.I. of the prism
(B) Will be 2D for a ray of R.I. 2.4 if it is D for a ray of R.I. 1.2
(C) Is directly proportional to the angle of the prism
(D) Will decrease with the increase in R.I. of a prism

14. A ray of light is incident from air to a glass rod at point A the angle of incidence being 45° . The minimum value of refractive index of the material of the rod so that T.I.R. takes place at B is



15. Refraction takes place at a concave spherical boundary separating glass air medium. For the image to be real, the object distance ($\mu_g = 3/2$):

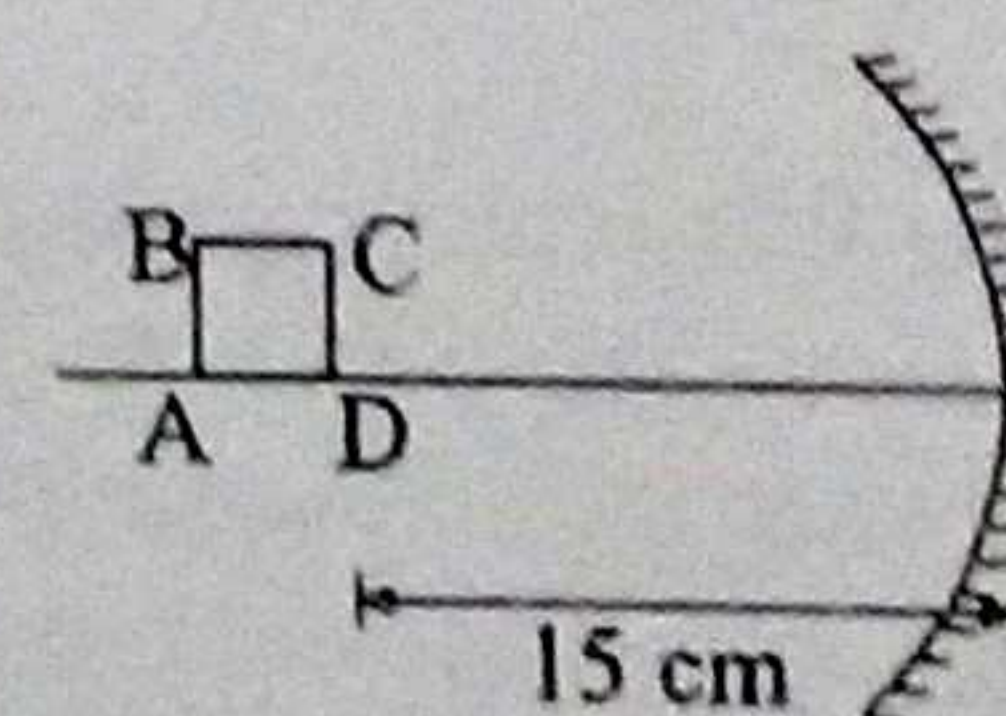
- (A) Should be greater than three times the radius of curvature of the refracting surface
(B) Should be greater than two times the radius of curvature of the refracting surface
(C) Should be greater than the radius of curvature of the refracting surface
(D) Is independent of the radius of curvature of the refracting surface

16. A point source of light at the surface of a sphere causes a parallel beam of light to emerge from the opposite surface of the sphere. The refractive index of the material of the sphere is :

- (A) 1.5 (B) 5/3 (C) 2 (D) 2.5

17. A square ABCD of side 1mm is kept at distance 15 cm in front of the concave mirror as shown in the figure. The focal length of the mirror is 10 cm. The length of the perimeter of its image will be :

- (A) 8 mm (B) 2 mm
(C) 12 mm (D) 6 mm

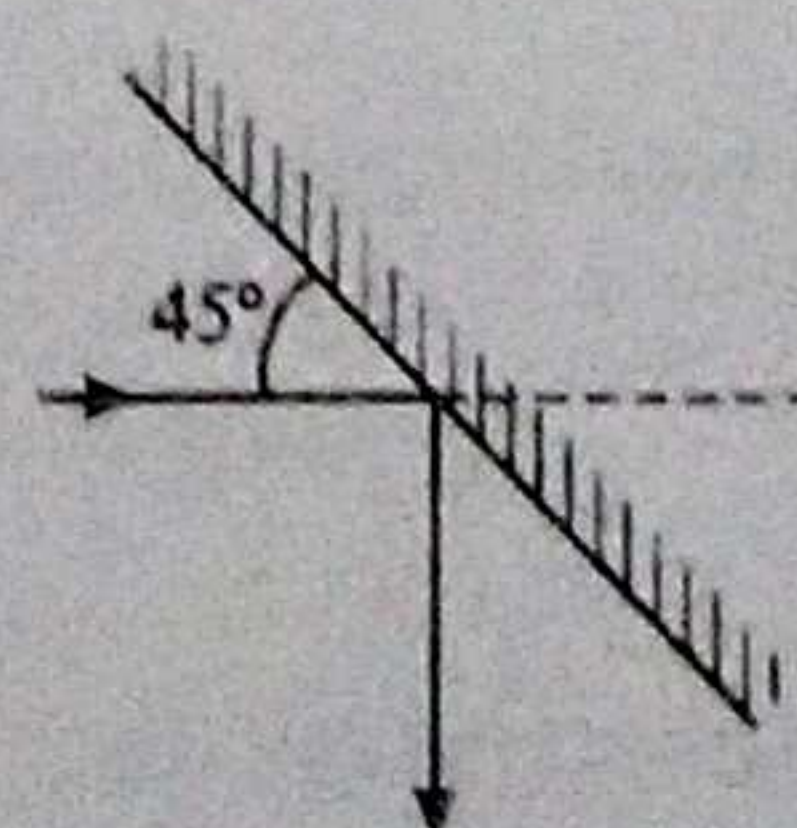
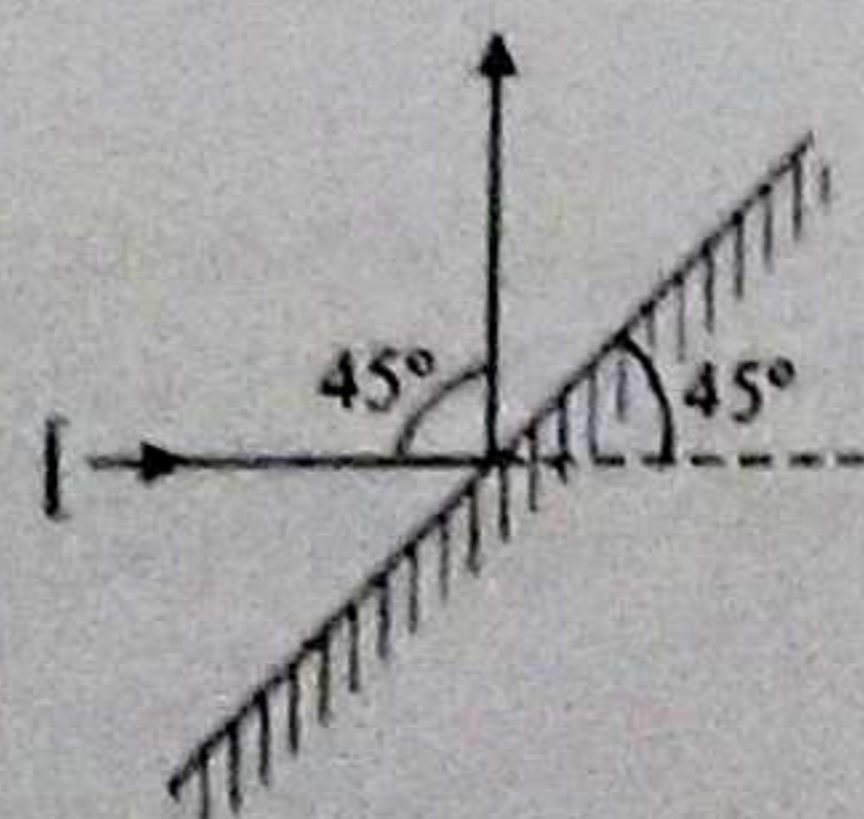


* * * * *

ANSWER KEY

1. (B) 2. (A)

3.



The plane mirror must be placed making an

- angle of 45° with horizontal facing towards the object.] 4. [120° anticlockwise and 240° clockwise] 5. (B)
6. [Infinitely large] 7. (C) 8. [30° clockwise] 9. (B) 10. (A)(C)(D) 11. (A)
12. (A)(D) 13. (A)(C) 14. $[\sqrt{3}/2]$ 15. (A) 16. (C) 17. (C)

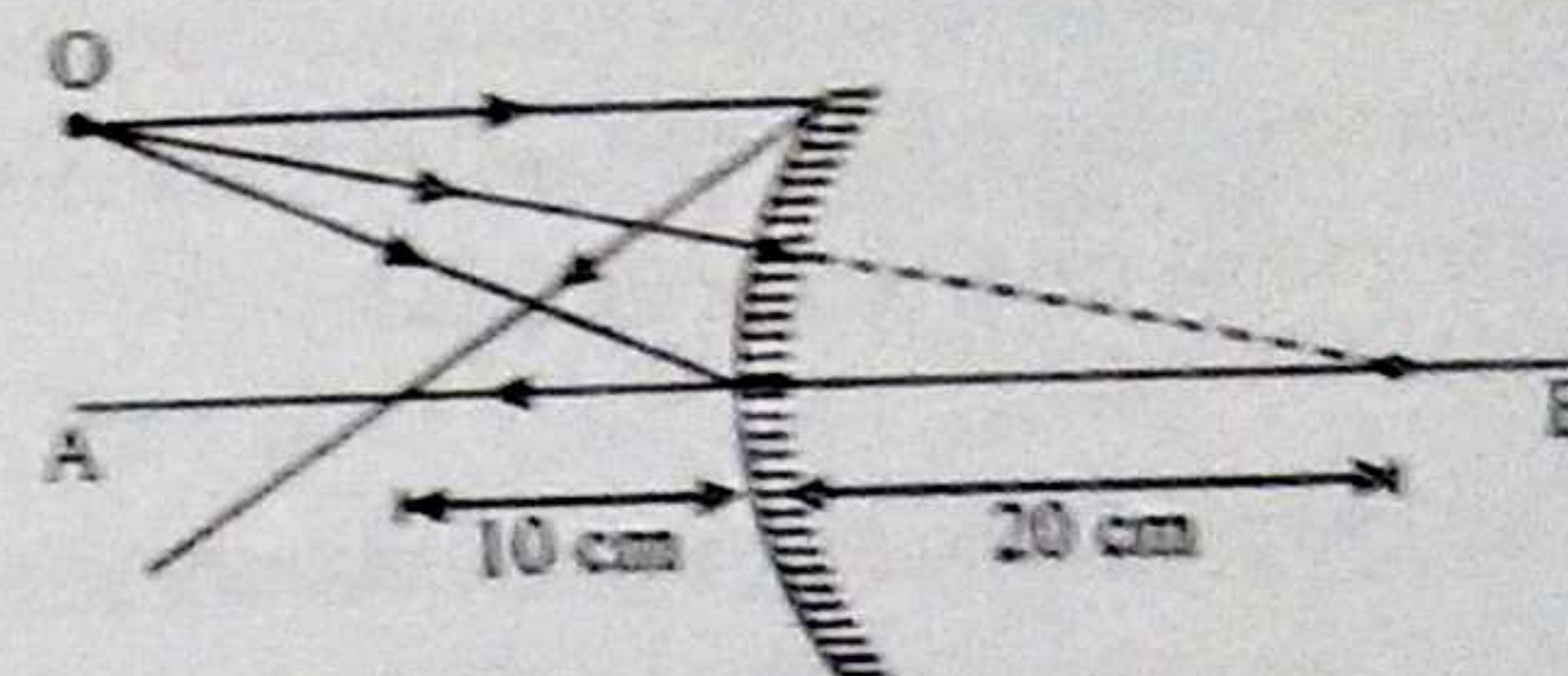
Geometrical Optics

Type of CPP - Objective & Subjective

Level - MODERATE

1. Find the diameter of the image of the moon formed by a spherical concave mirror of focal length 11.4 m. The diameter of the moon is 3450 km and the distance between the earth and the moon is 3.8×10^5 km.

2. In the figure shown a convex mirror of a radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the ray diagram (AB is its principal axis):
 (A) 3 (B) 2
 (C) 1 (D) 0



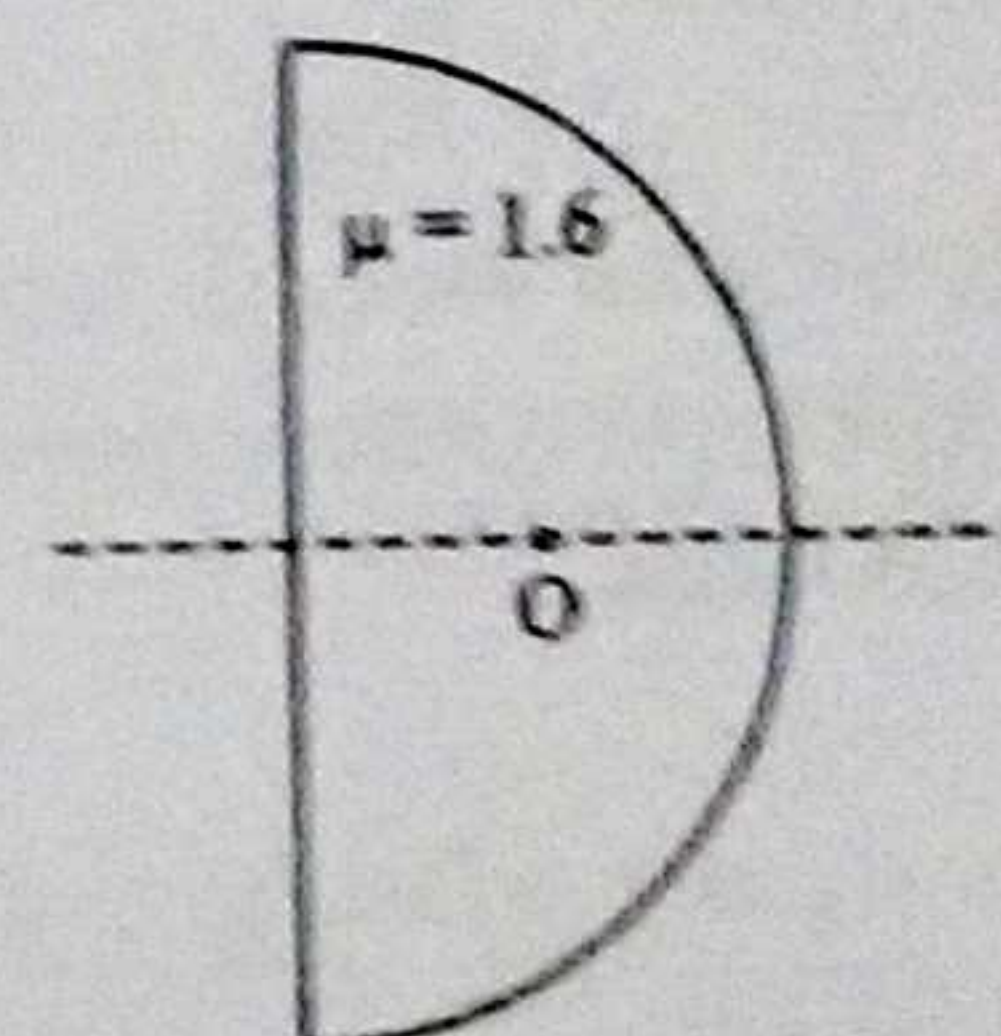
3. The radius of curvature of a convex spherical mirror is 1.2 m. How far away from the mirror is an object of height 12 cm if the distance between its virtual image & the mirror is 0.35 m? What is the height of the image?

4. The sun (diameter D) subtends an angle θ radian at the pole of a concave mirror of focal length f . The diameter of the image of the sun formed by the mirror is:

(A) $f\theta$ (B) $2f\theta$ (C) $\frac{2f\theta}{D}$ (D) $D\theta$

5. A plastic hemisphere has a radius of curvature of 8 cm and an index of refraction of 1.6. On the axis halfway between the plane surface and the spherical one (4 cm from each) is a small object O . The distance between the two images when viewed along the axis from the two sides of the hemisphere is approximately:

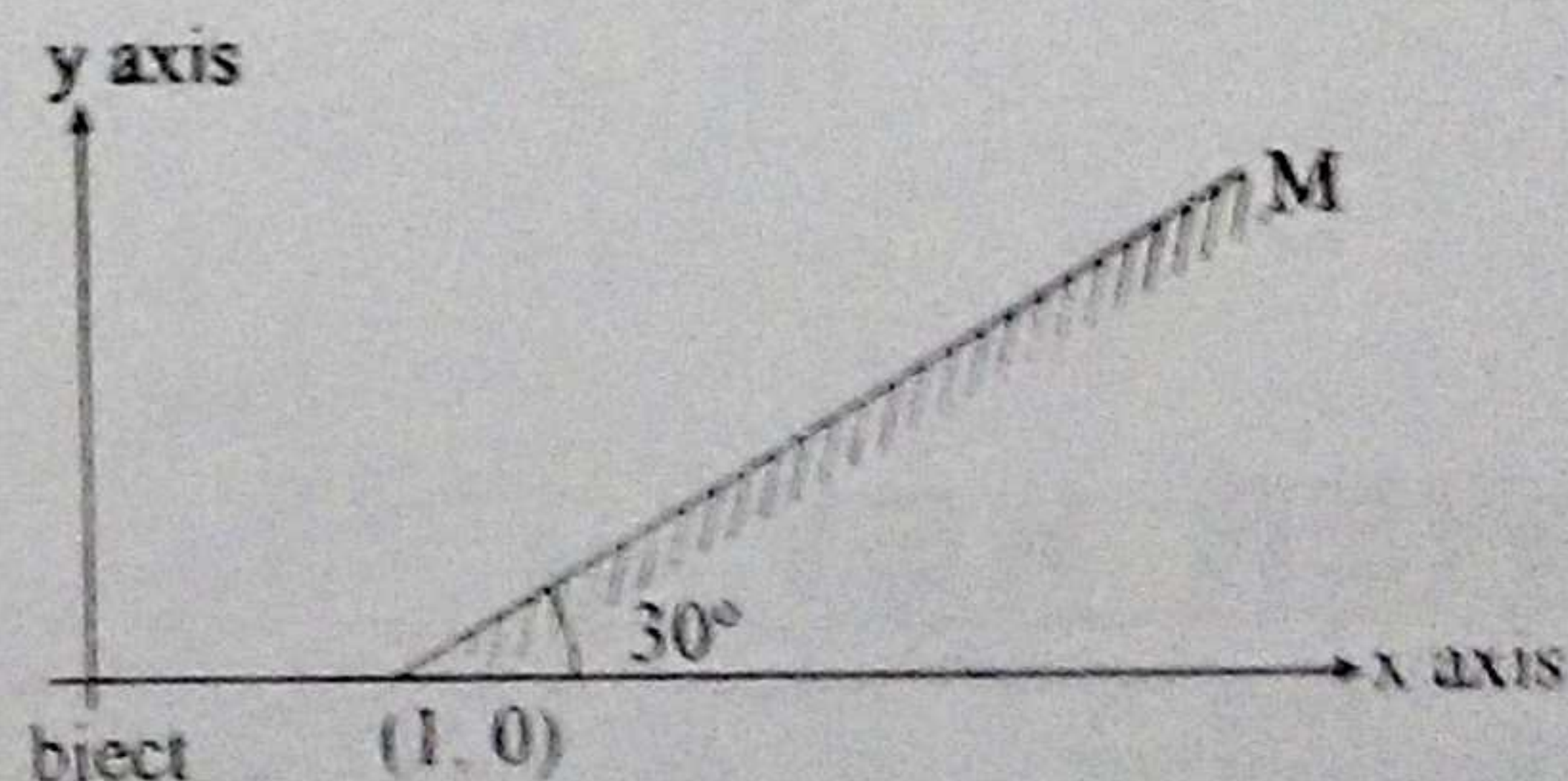
(A) 1.0 cm (B) 1.5 cm
 (C) 3.75 cm (D) 2.5 cm



6. Sun rays are incident at an angle of 24° with the horizon. How can they be directed parallel to the horizon using a plane mirror?

7. A person's eye is at a height of 1.5 m. He stands in front of a 0.3 m long plane mirror which is 0.8 m above the ground. The length of the image he sees of himself is:
 (A) 1.5 m (B) 1.0 m (C) 0.8 m (D) 0.6 m

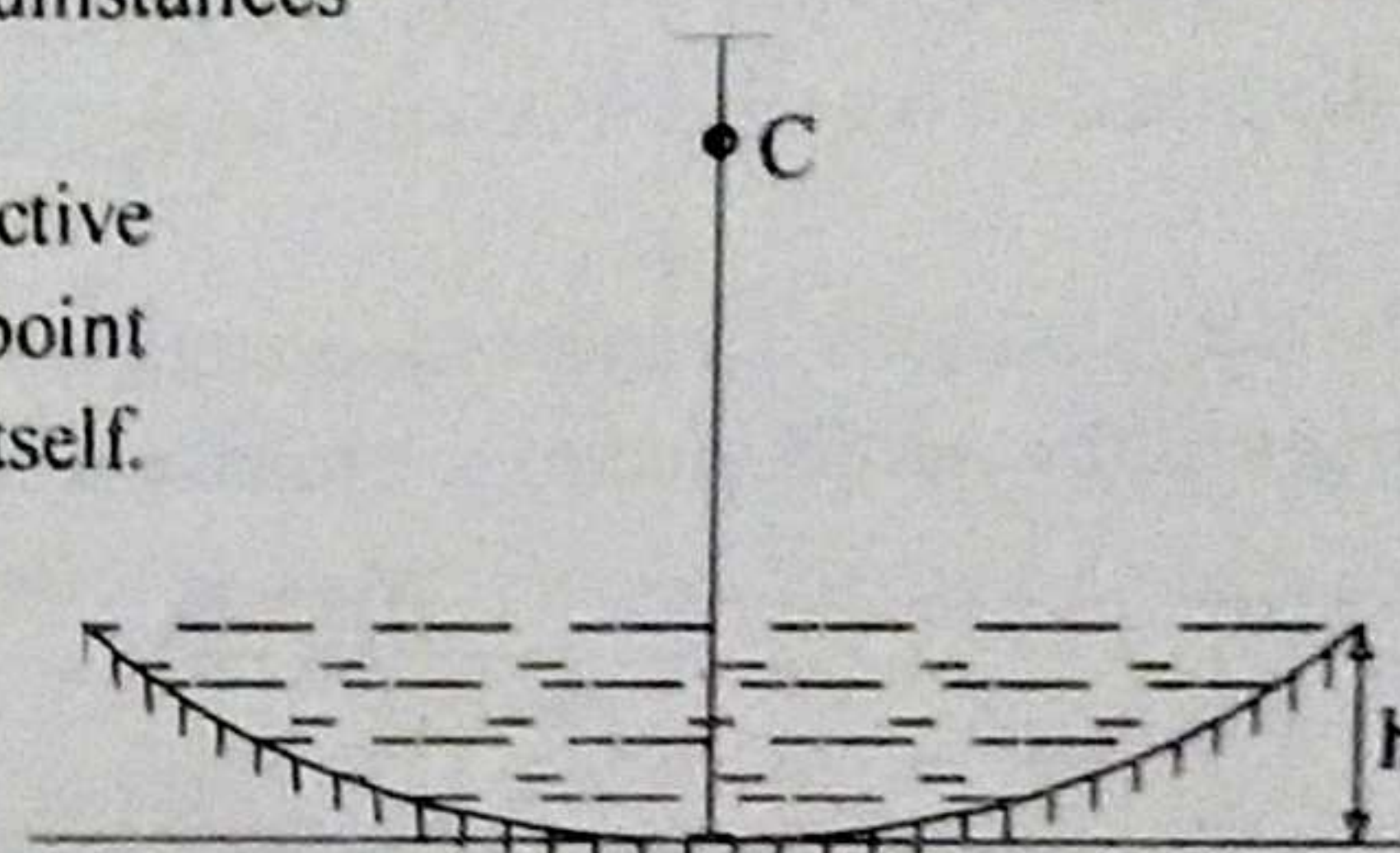
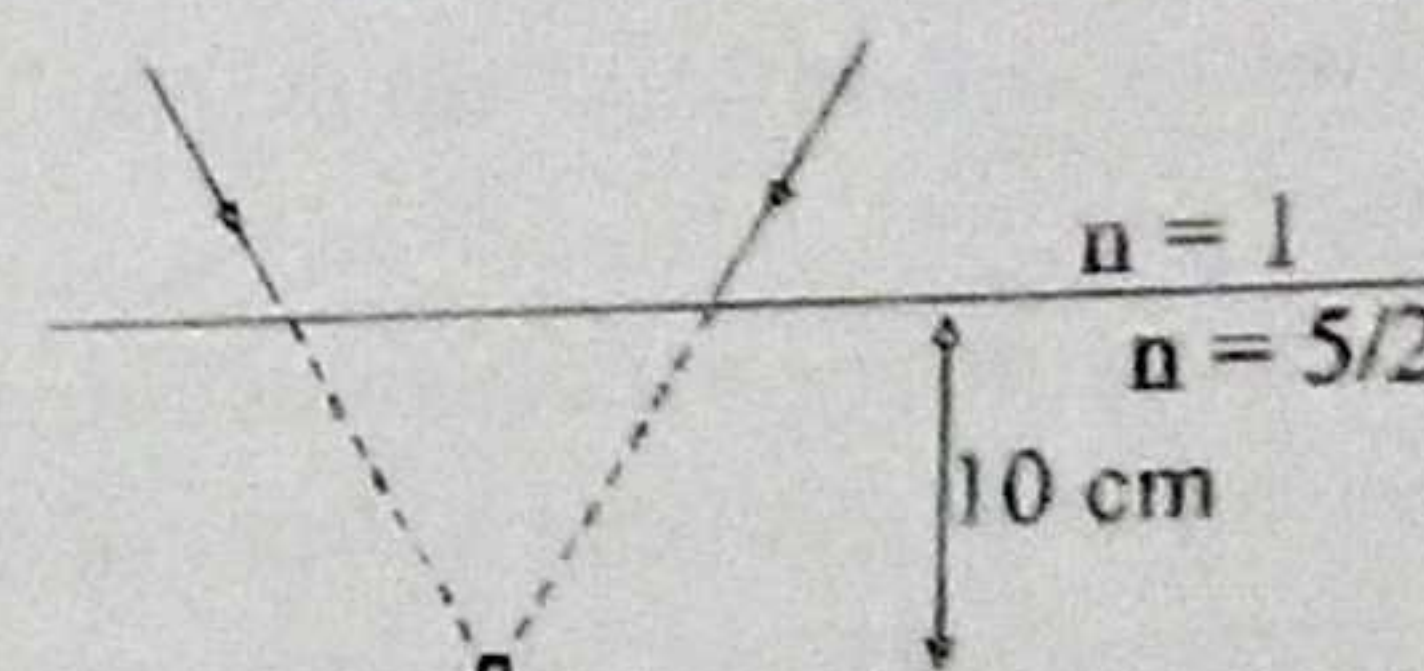
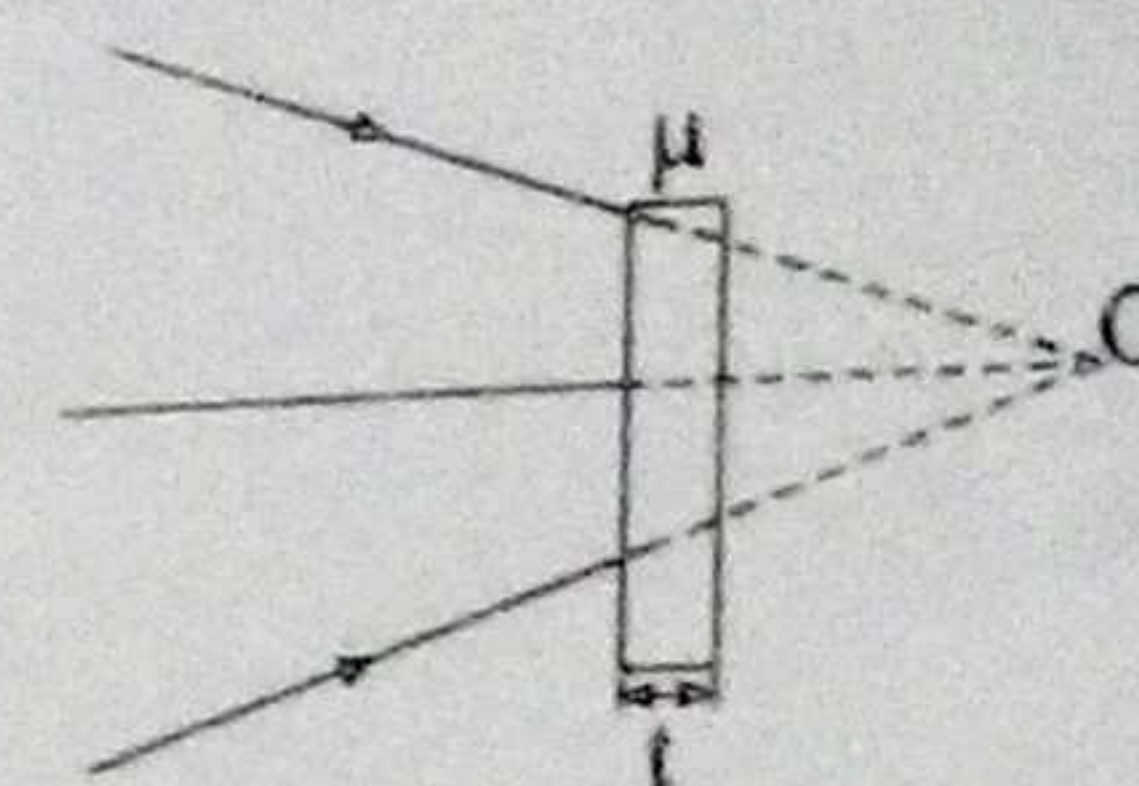
8. A point object is placed at $(0, 0)$ and a plane mirror M is placed, inclined 30° with the x axis.
 (a) Find the position of image.
 (b) If the object starts moving with velocity $1 \hat{i}$ m/s and the mirror is fixed find the velocity of image.



9. Two parallel rays are travelling in a medium of refractive index $\mu_1 = 4/3$. One of the rays passes through a parallel glass slab of thickness t and refractive index $\mu_2 = 3/2$. The path difference between the two rays due to the glass slab will be:
 (A) $4t/3$ (B) $3t/2$ (C) $t/8$ (D) $t/6$

10. A ray of light is incident on a glass sphere of refractive index $3/2$. What should be the angle of incidence so that the ray which enters the sphere does not come out of the sphere:
 (A) $\tan^{-1}(2/3)$ (B) $\sin^{-1}(2/3)$ (C) 90° (D) $\cos^{-1}(1/3)$

11. A beam of light is converging towards a point. A plane parallel plate of glass of thickness t refractive index μ is introduced in the path of the beam. The convergent point is shifted by (assume near normal incidence) :
- (A) $t\left(1 - \frac{1}{\mu}\right)$ away (B) $t\left(1 + \frac{1}{\mu}\right)$ away
(C) $t\left(1 - \frac{1}{\mu}\right)$ nearer (D) $t\left(1 + \frac{1}{\mu}\right)$ nearer
12. When a wave is refracted :
(A) Its path must change (B) Its amplitude must change
(C) Its velocity must change (D) Its frequency must change
13. Rays incident on an interface would converge 10 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet some where else. Find the distance of meeting point of refracted rays below the interface, assuming the rays to be making small angles with the normal to the interface.
14. The light reflected by a plane mirror may form a real image :
(A) If the rays incident on the mirror are converging (B) If the rays incident on the mirror are diverging
(C) If the object is placed very close to the mirror (D) Under no circumstances
15. A concave mirror of radius R is kept on a horizontal table (figure). Water (refractive index = μ) is poured into it upto a height h . What should be distance of a point object from surface along principal axis so that its final image is formed on itself. Consider two cases.
(i) $h \rightarrow 0$
(ii) in terms of h
16. How much water should be filled in a container of 21 cm in height, so that it appears half filled (of total height of the container) when viewed from the top of the container ? (Assume near normal incidence and $\mu_w = 4/3$)
(A) 8.0 cm (B) 10.5 cm (C) 12.0 cm (D) 14.0 cm
17. For a glass prism ($\mu = \sqrt{2}$) the angle of minimum deviation is equal to the refracting angle of the prism. The angle of the prism is :
(A) 80° (B) 45° (C) 60° (D) 90°



ANSWER KEY

1. $[10.35 \text{ cm} = \frac{3933}{380} \text{ cm}]$ 2. (B) 3. $[84 \text{ cm}, 0.05 \text{ m}]$ 4. (A) 5. (D)
6. [Mirror should be placed on the path of the rays at an $\angle$ of 78° or 12° to the horizontal] 7. (D)
8. [(a) Position of image = $(1 \cos 60^\circ, -1 \sin 60^\circ)$; (b) Velocity of image = $(1 \cos 60^\circ \hat{i} + 1 \sin 60^\circ \hat{j}) \text{ m/s}$] 9. (C)
10. (C) 11. (A) 12. (B) 13. $[25 \text{ cm}]$ 14. (A) 15. [(i) $\frac{R}{\mu}$; (ii) $\frac{(R-h)}{\mu}$] 16. (D) 17. (D)

Geometrical Optics

Type of CPP - Objective & Subjective

Level - MODERATE

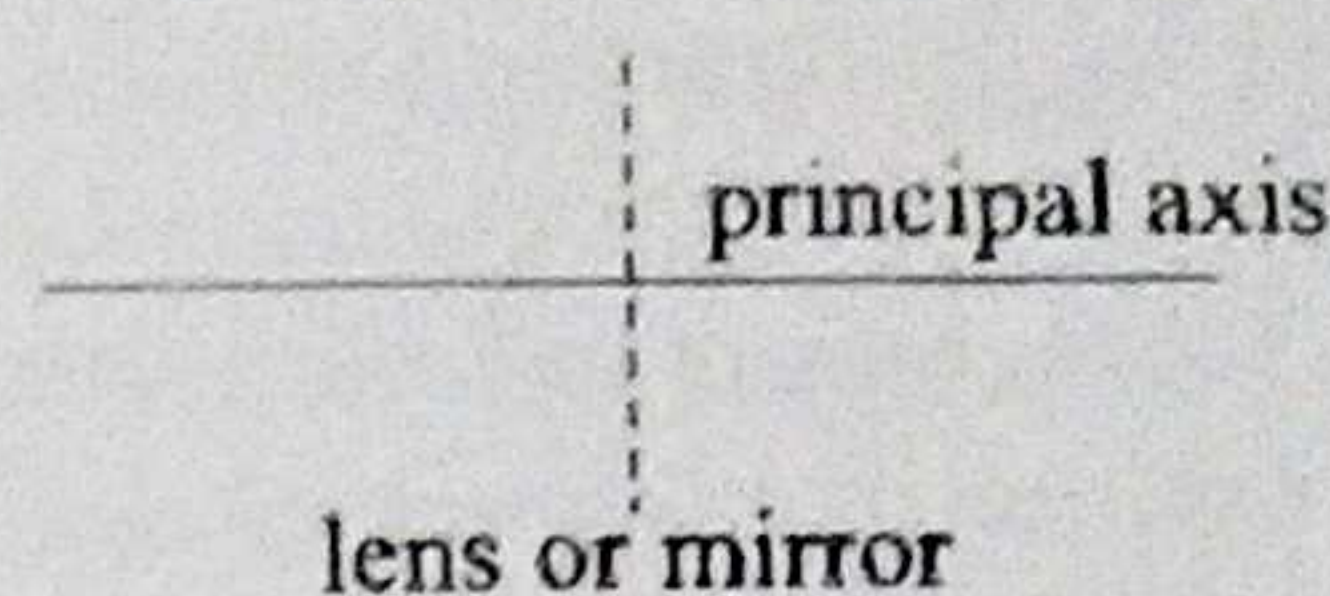
1. Column-I gives certain situations regarding a point object and its image formed by an optical instrument. The possible optical instruments are concave and convex mirrors or lenses as given in Column-II. Same side of principal axis means both image and object should either be above the principal axis or both should be below the principal axis as shown in figure. Same side of optical instrument means both image and object should be either left of the optical instrument or both should be on right of the optical instrument as shown in figure. Match the statements in column-I with the corresponding statements in column-II :

Column-I

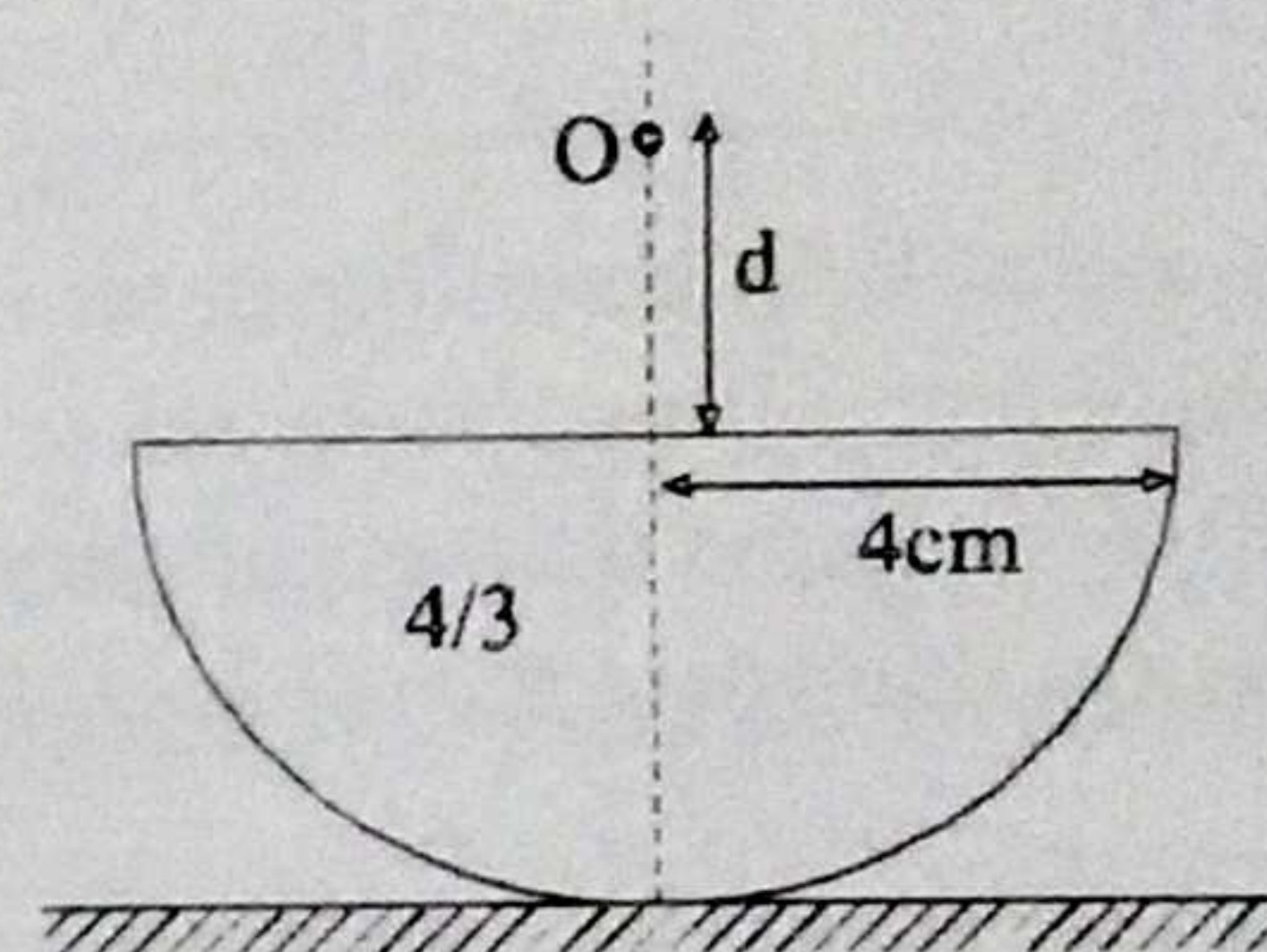
- (A) If point object and its image are on same side of principal axis and opposite sides of the optical instrument then the optical instrument is
(B) If point object and its image are on opposite side of principal axis and same sides of the optical instrument then the optical instrument is
(C) If point object and its image are on same side of principal axis and same sides of the optical instrument then the optical instrument is
(D) If point object and its image are on opposite side of principal axis and opposite sides of the optical instrument then the optical instrument

Column-II

- (p) Concave mirror
(q) Convex mirror
(r) Concave lens
(s) Convex lens

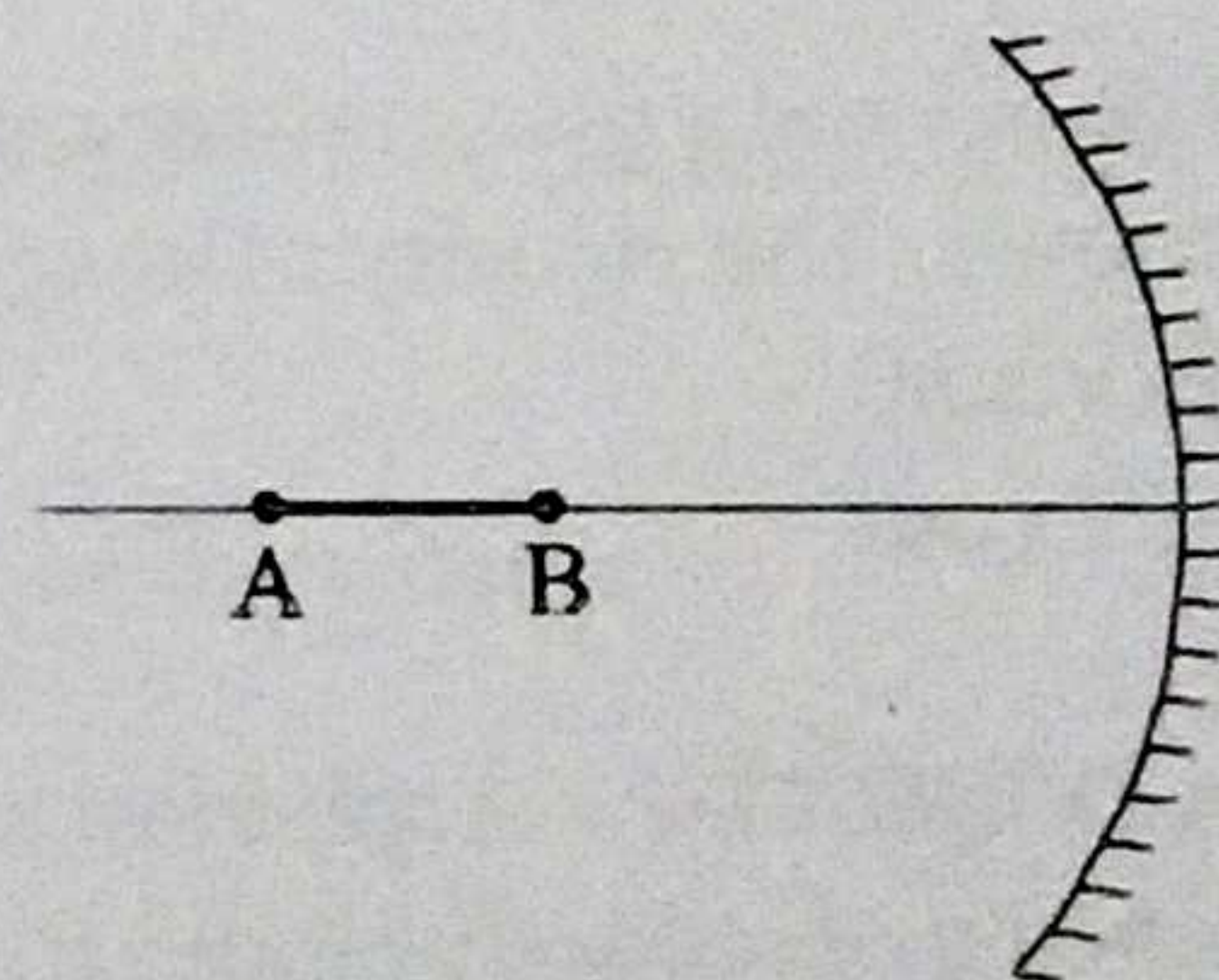


2. A glass hemisphere of refractive index $4/3$ and of radius 4 cm is placed on a plane mirror. A point object is placed on axis of this sphere at a distance ' d ' from O as shown. If the final image is formed at infinity, then find the value of ' d ' in cm .



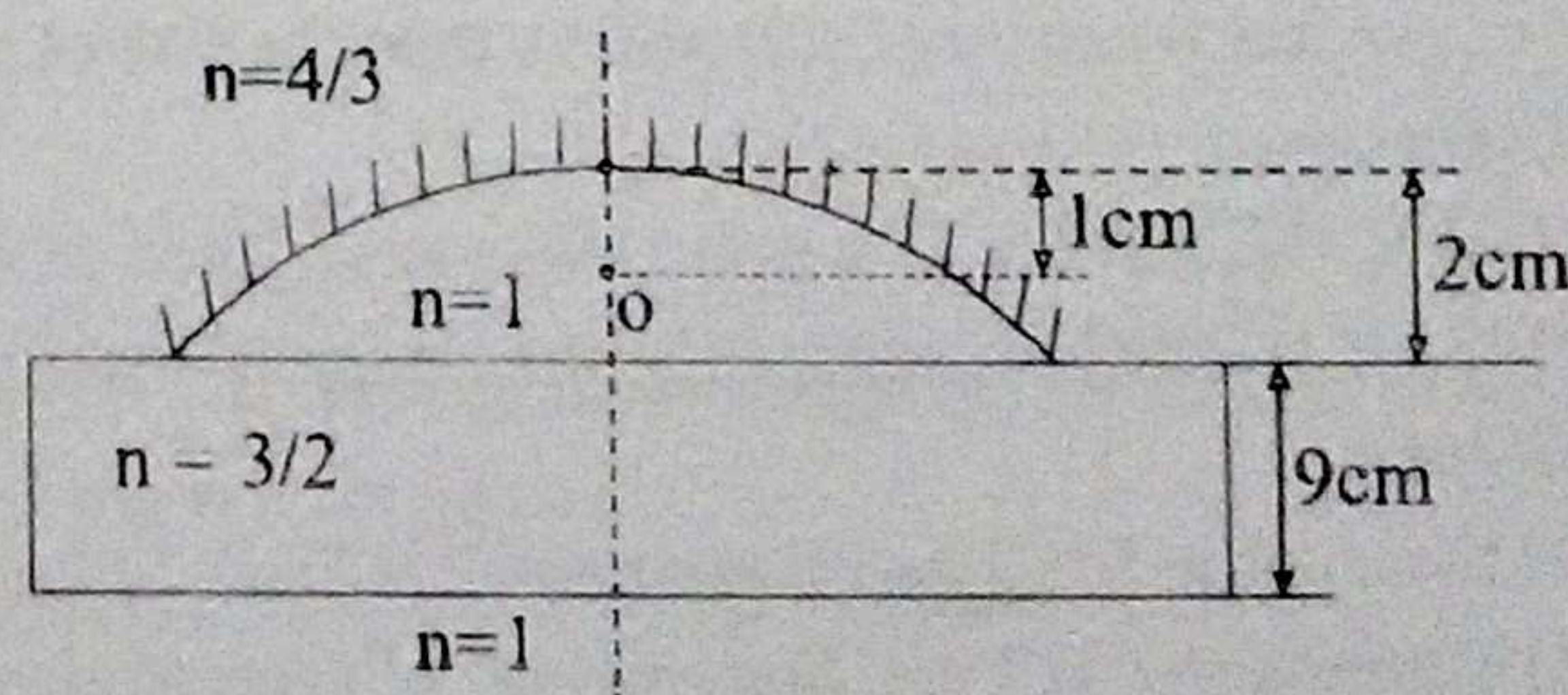
3. A linear object AB is placed along the axis of a concave mirror. The object is moving towards the mirror with speed V . The speed of the image of the point A is $4V$ and the speed of the image of B is also $4V$. If centre of the line AB is at a distance L from the mirror then length of the object AB will be :

- (A) $\frac{3L}{2}$ (B) $\frac{5L}{3}$
(C) L (D) $\frac{4L}{3}$



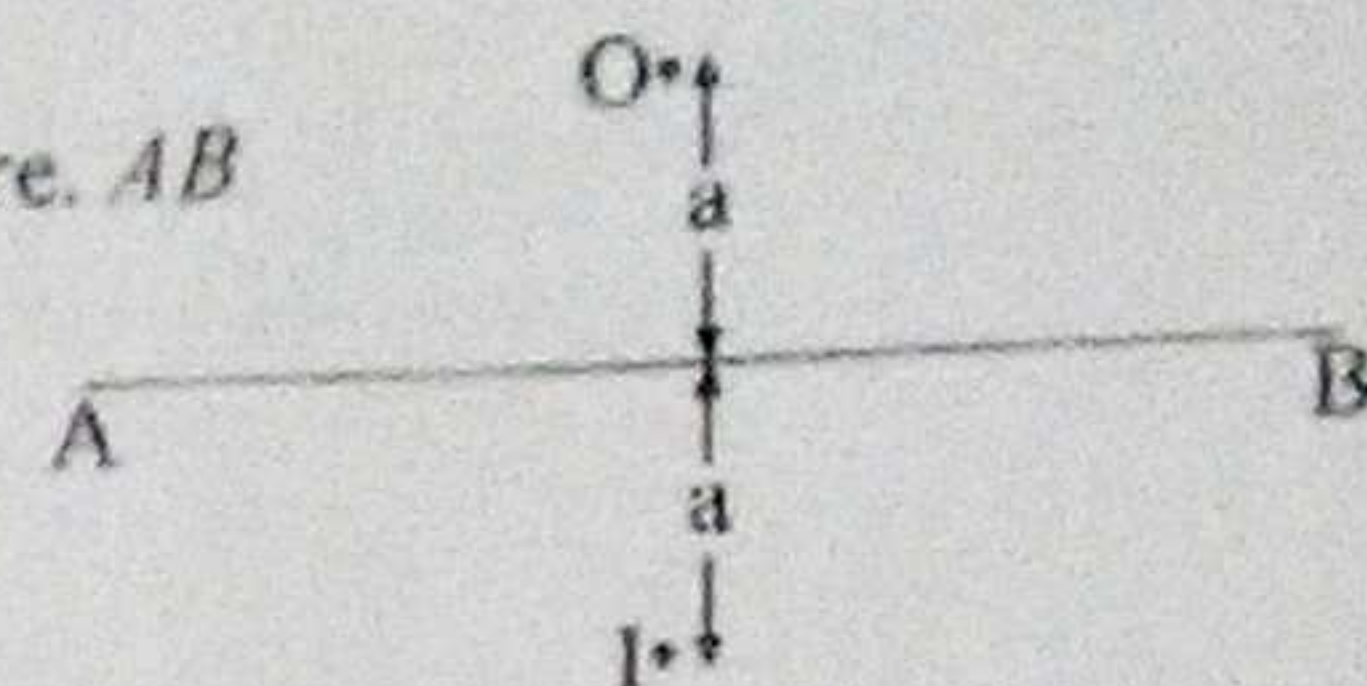
4. A concave mirror of focal length 2 cm is placed on a glass slab as shown in the figure. Then the image of object O formed due to reflection at mirror and then refraction by the slab :

- (A) Will be virtual and will be at 2 cm from the pole of the concave mirror
(B) Will be virtual and formed on the pole of the mirror
(C) Will be real and on the object itself
(D) None of these



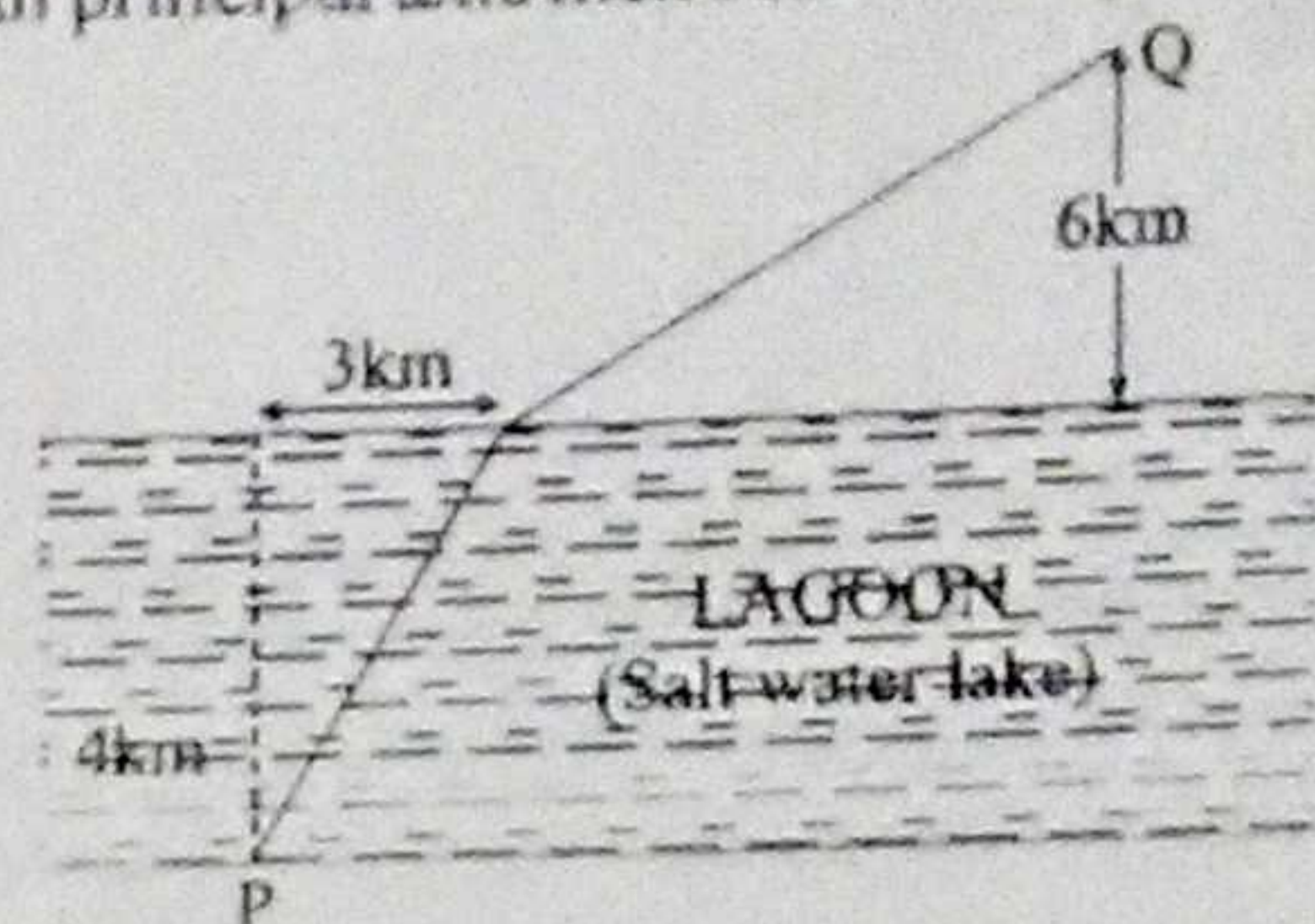
5. A thin prism of angle 6.0° , $\omega = 0.07$ and $\mu_y = 1.50$ is combined with another thin prism having $\omega = 0.08$ and $\mu_y = 1.60$. The combination produces no deviation in the mean ray. (a) Find the angle of the second prism. (b) Find the net angular dispersion produced by the combination when a beam of white light passes through it. (c) If the prisms are similarly directed, what will be the deviation in the mean ray? (d) Find the angular dispersion in the situation described in (c).

6. The position of a real point object and its real point image are as shown in the figure. AB is the principal axis. This can be achieved by using :
 (A) Convex mirror (B) Concave mirror
 (C) Concave mirror only (D) Convex mirror only



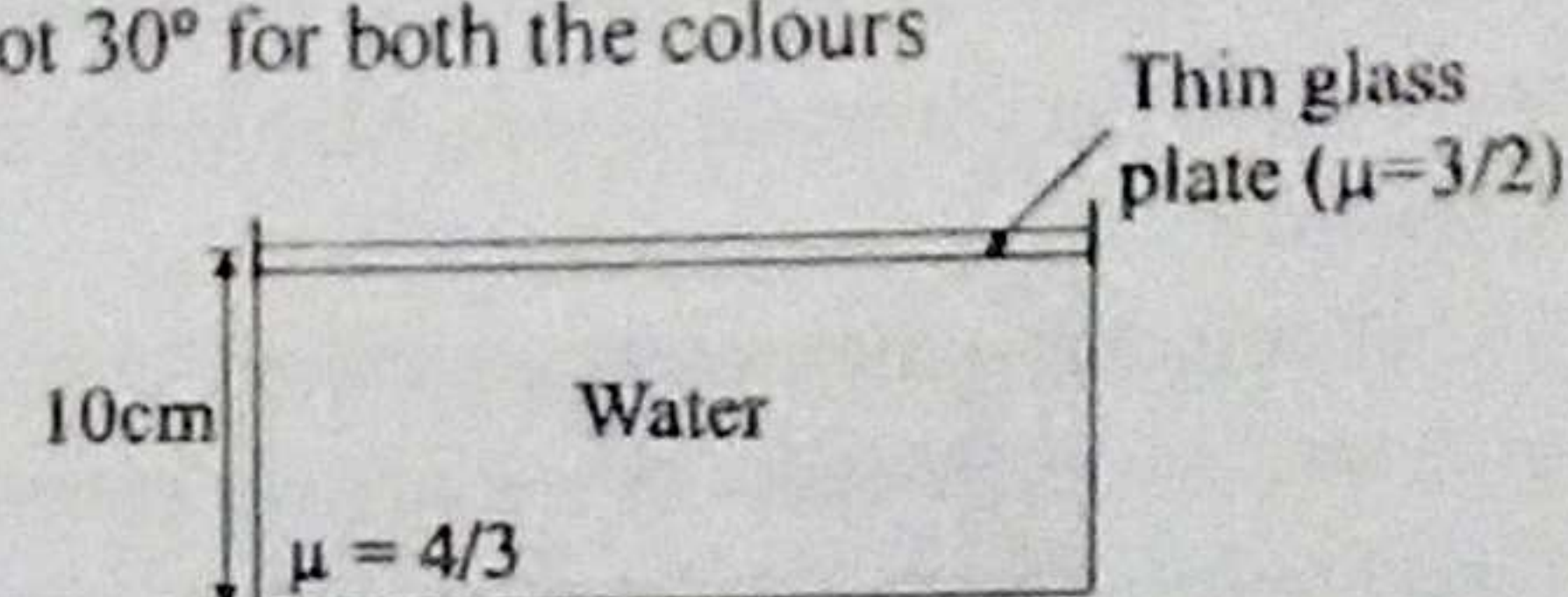
7. **State true 'or' false :** A ray is incident on spherical mirror at an angle $\theta = 30^\circ$ with principal axis then reflected ray must make an angle $\theta = 30^\circ$ with principal axis.

8. A man starting from point P crosses a 4 km wide lagoon and reaches point Q in the shortest possible time by the path shown. If the person swims at a speed of 3 km/hr and walks at a speed of 4 km/hr, then his time of journey (in minutes) is ?



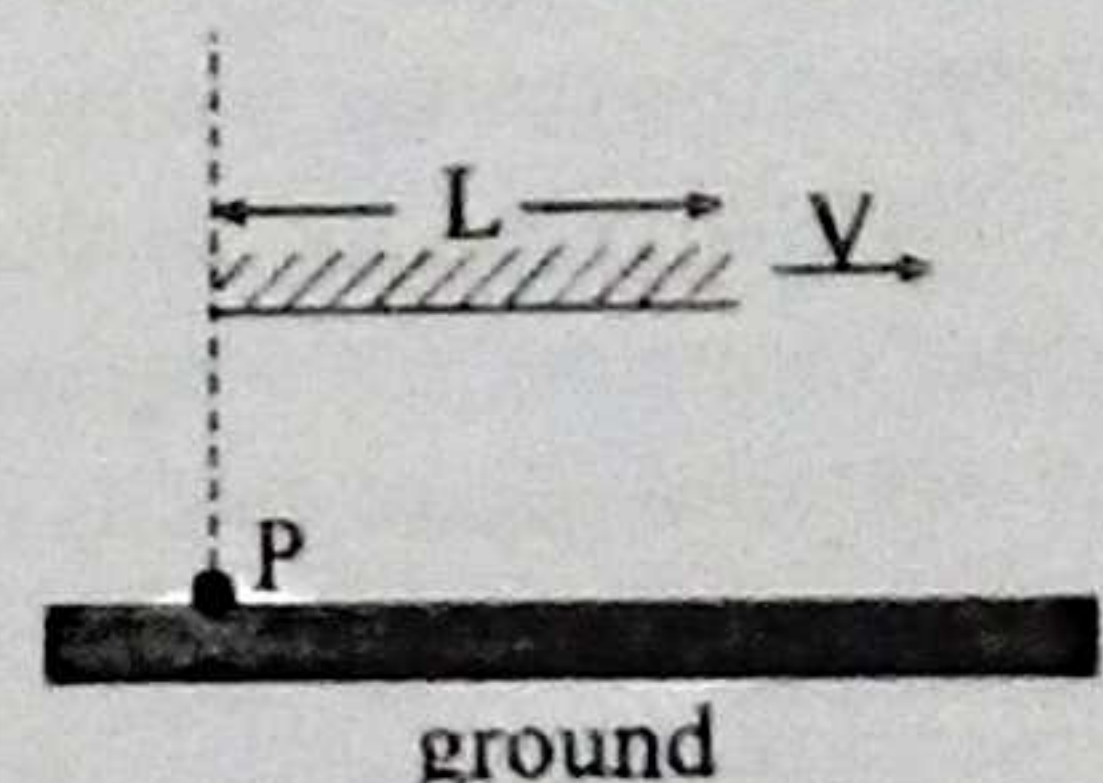
9. Two beams of red and violet colours are made to pass separately through a prism (angle of the prism is 60°). In the position of minimum deviation, the angle of refraction will be :
 (A) 30° for both the colours (B) Greater for the violet colour
 (C) Greater for the red colour (D) Equal but not 30° for both the colours

10. If the observer sees the bottom of vessel at 8 cm, find the refractive index of the medium in which observer is present.

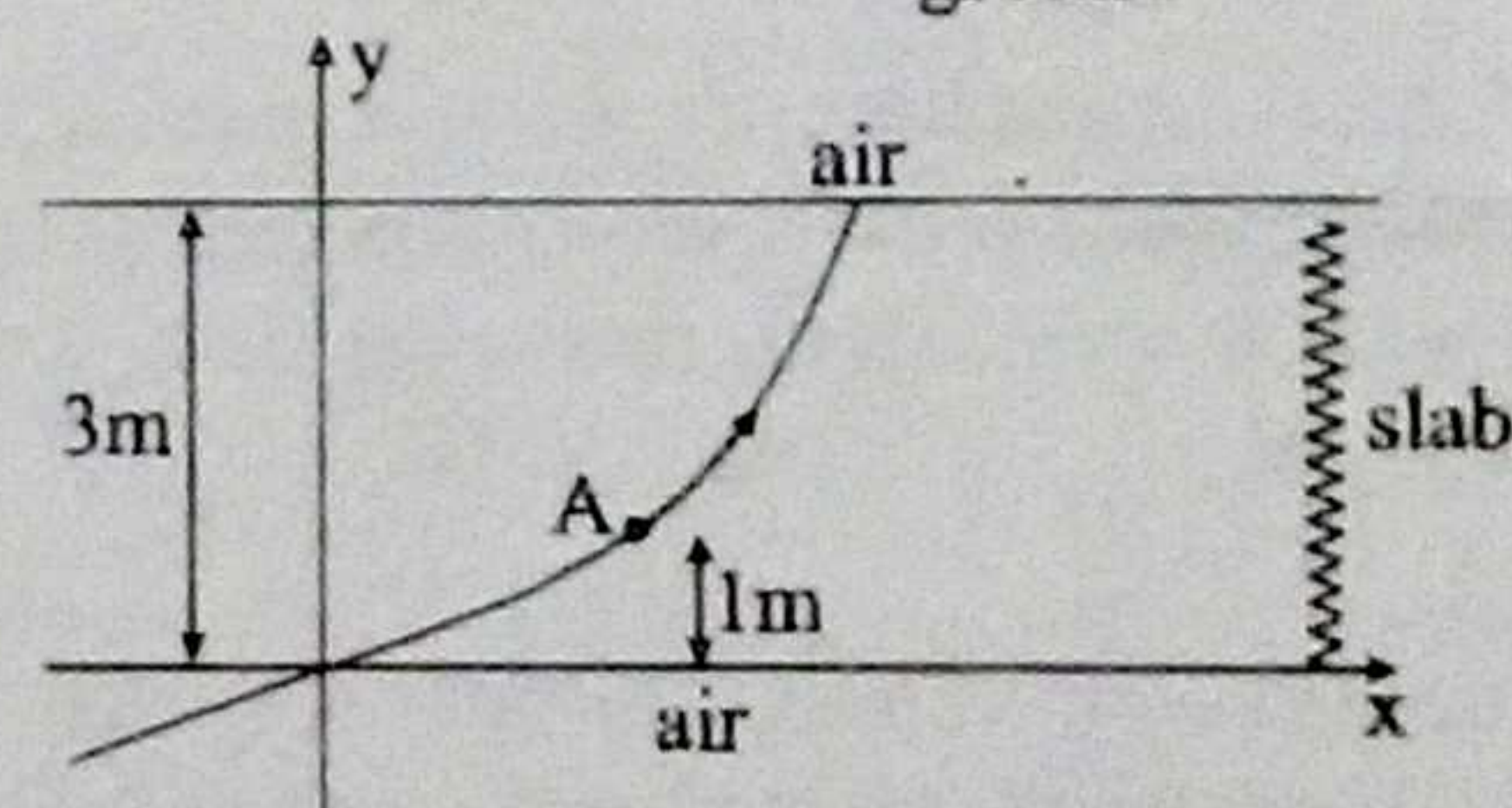


11. A mirror of length L moves horizontally as shown in the figure with a velocity v . The mirror is illuminated by a point source of light ' P ' placed on the ground. The rate at which the length of the light spot on the ground increases is :

- (A) v (B) Zero
 (C) $2v$ (D) $3v$

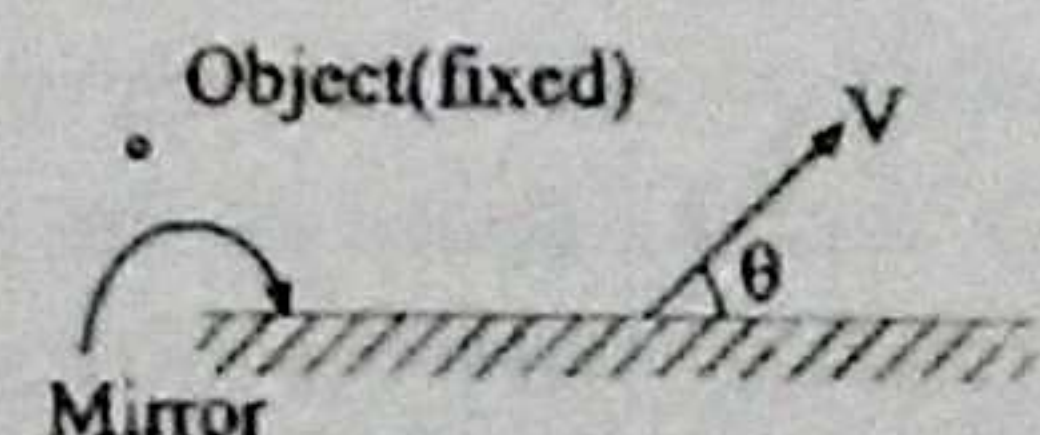


12. A ray of light travelling in air is incident at grazing angle on a slab of 3m thickness with variable refractive index, $n(y) = [ky^{3/2} + 1]^{1/2}$ where $k = 1 \text{ m}^{-3/2}$ and follows path as shown. What is the total deviation produced by the slab when the ray is at A ($y = 1\text{m}$), in degrees ?

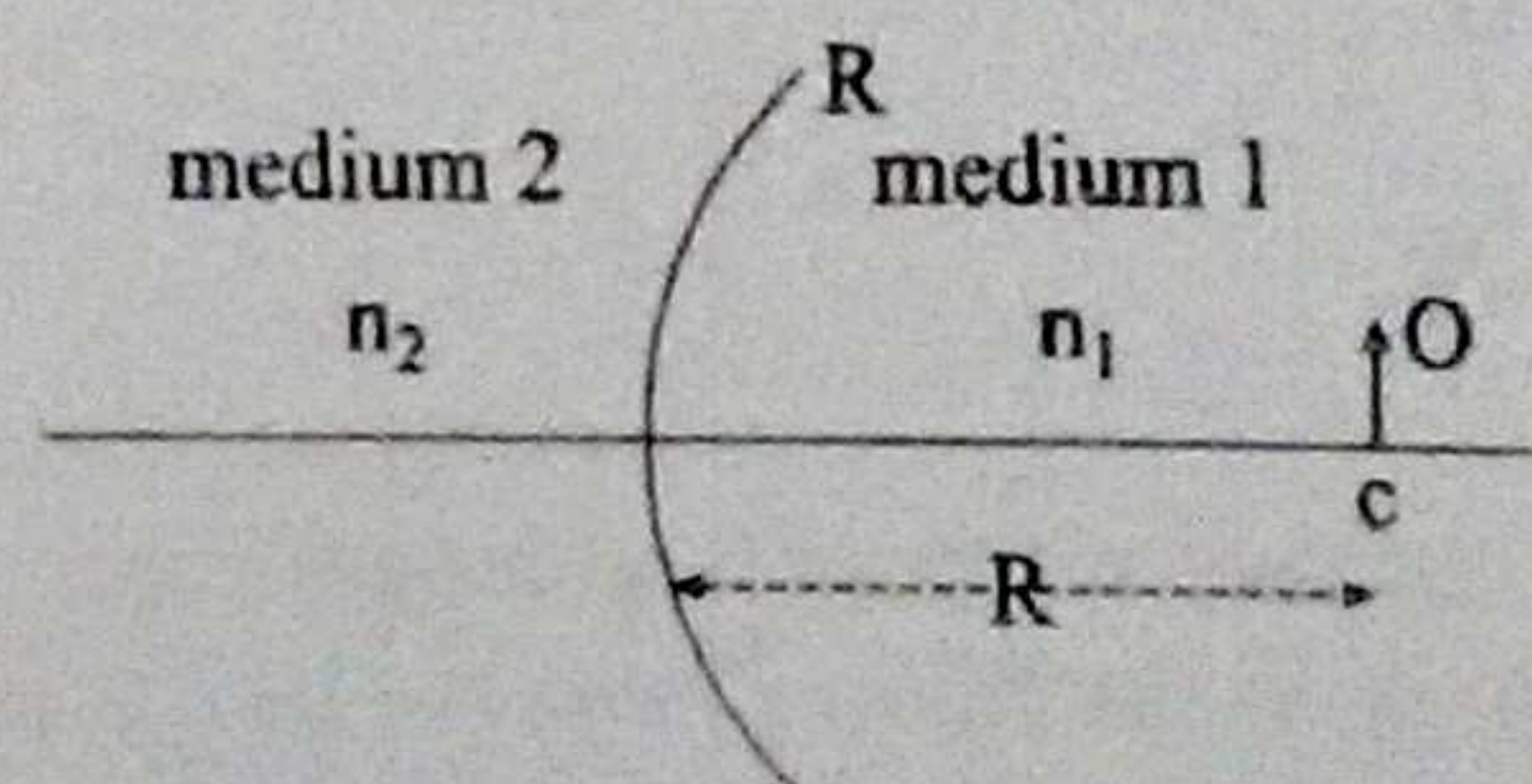


13. An object and a plane mirror are as shown in figure. Mirror is moved with velocity V as shown. The velocity of image is :

- (A) $2V \sin \theta$ (B) $2V$
 (C) $2V \cos \theta$ (D) None of these



14. **STATEMENT-1 :** A spherical surface of radius of curvature R separates two media of refractive index n_1 and n_2 as shown. If an object O (a thin small rod) is placed upright on principal axis at a distance R from pole (i.e., placed at centre of curvature), then the size of image is same as size of object.



STATEMENT-2 : If a point object is placed at centre of curvature of spherical surface separating two media of different refractive index, then the image is also formed at centre of curvature, i.e., image distance is equal to object distance.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False (D) Statement-1 is False, Statement-2 is True

ANSWER KEY

* * * * *

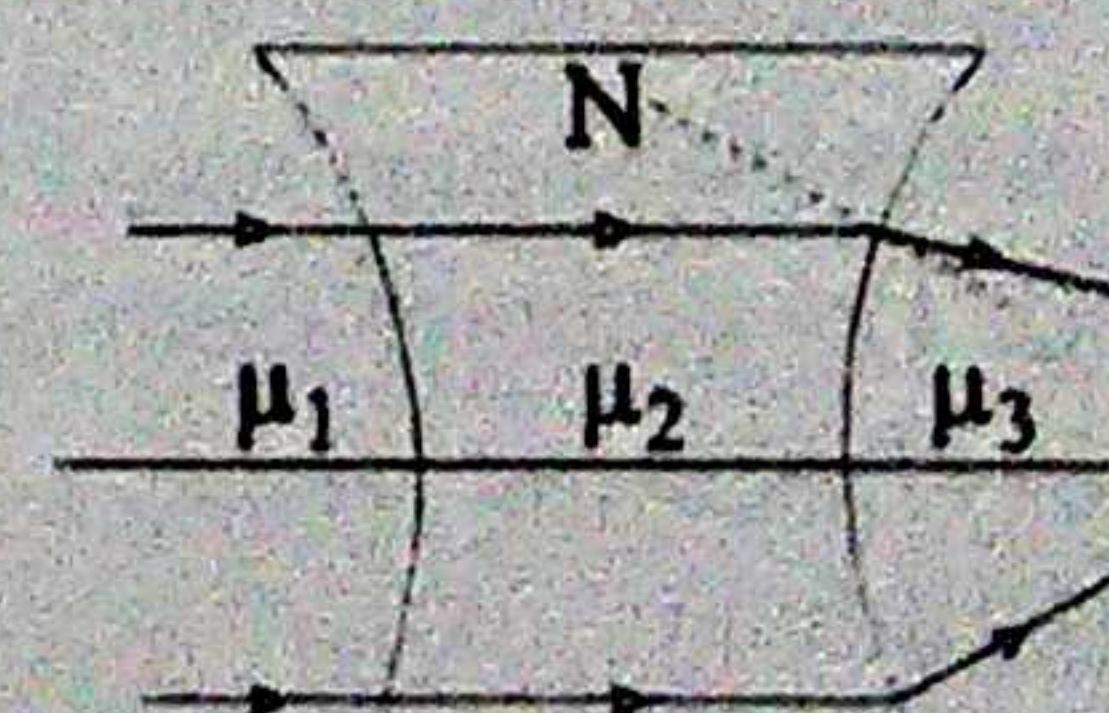
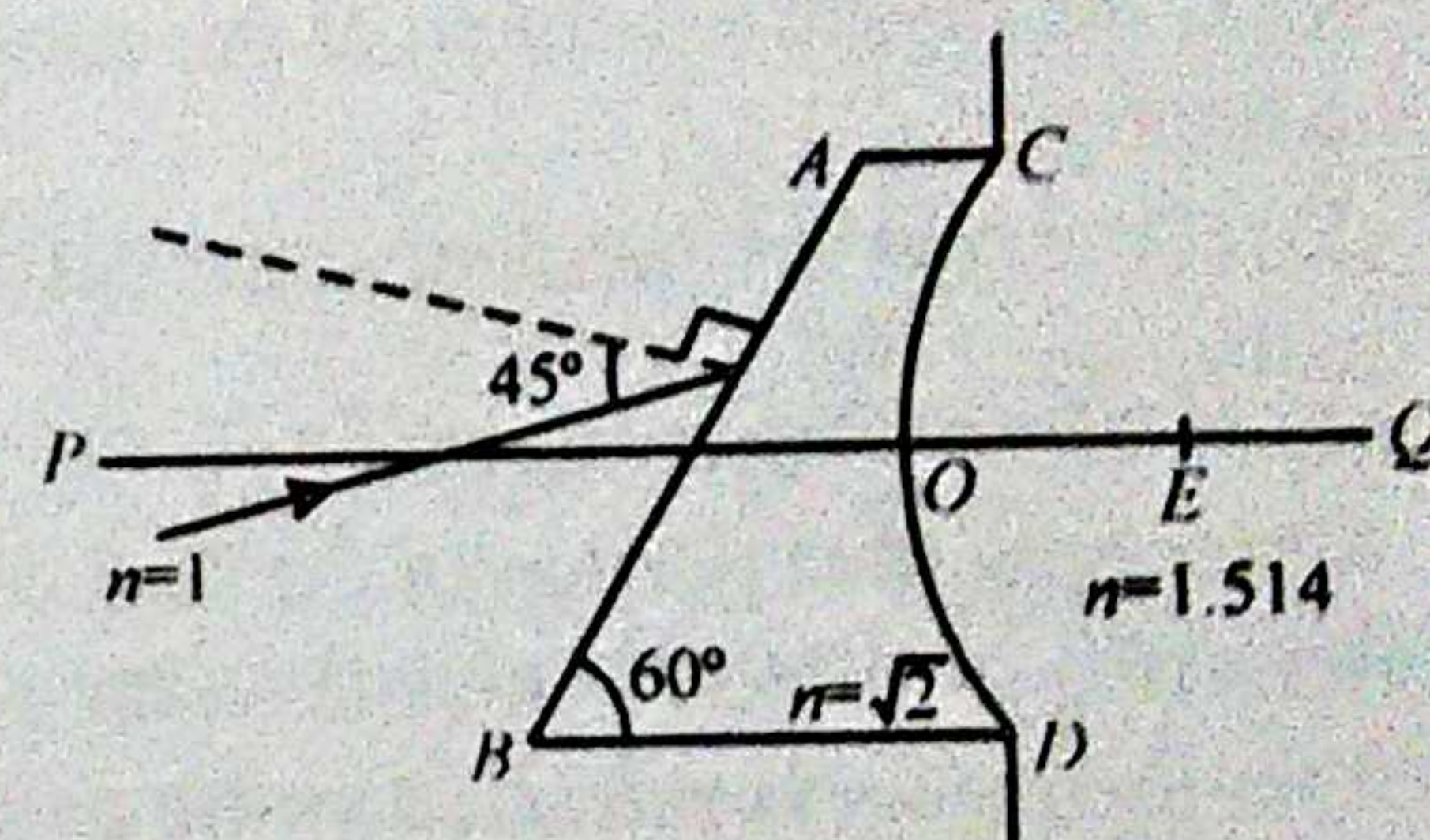
1. [(A) p, q (B) p, q (C) r, s (D) r, s] 2. [3 cm] 3. (C) 4. (D) 5. [(a) 5° ; (b) 0.03° ; (c) 6° ; (d) 0.45°] 6. (C)
 7. [False] 8. [250] 9. (A) 10. [$\frac{16}{15}$] 11. (B) 12. [45] 13. (A) 14. (D)

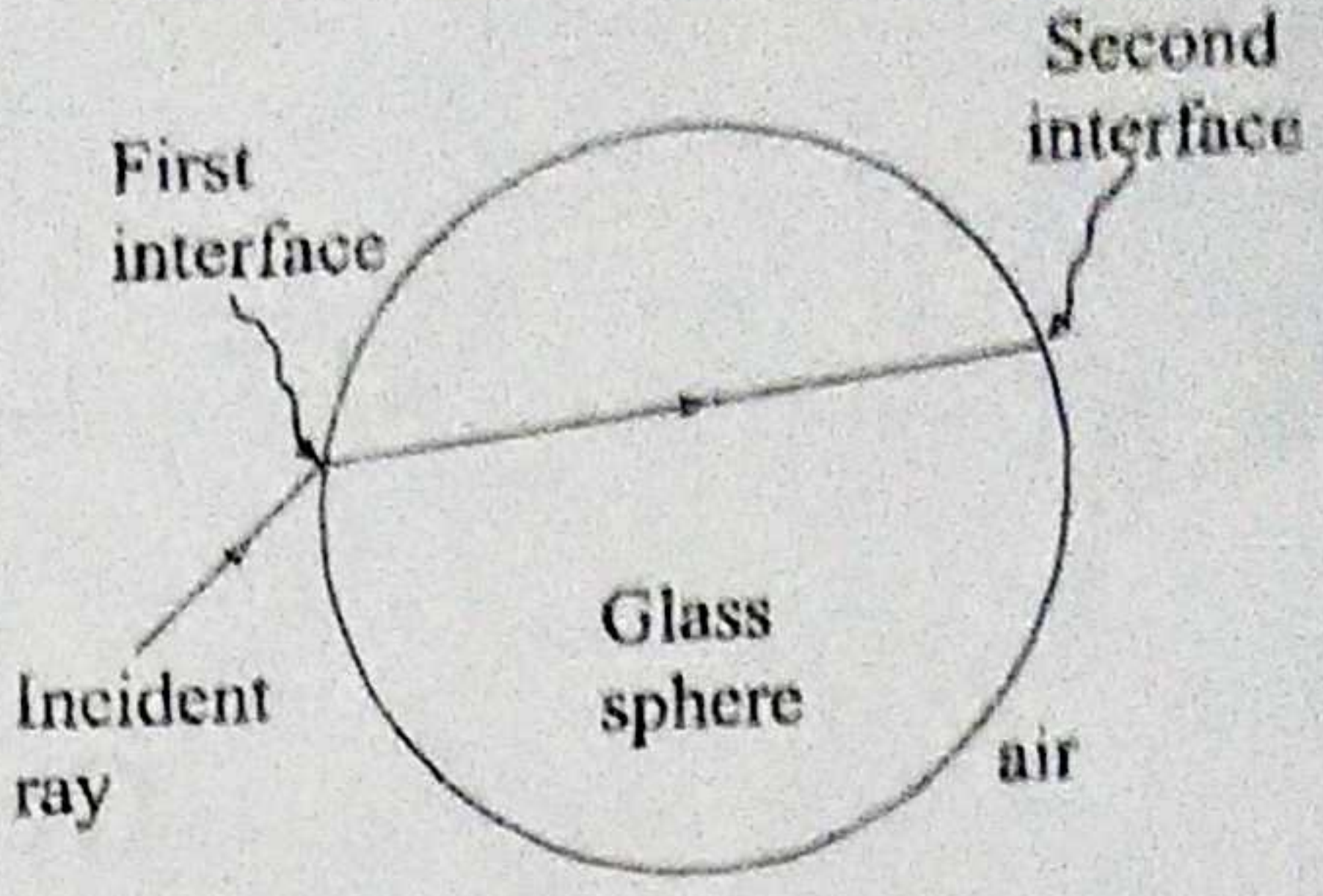
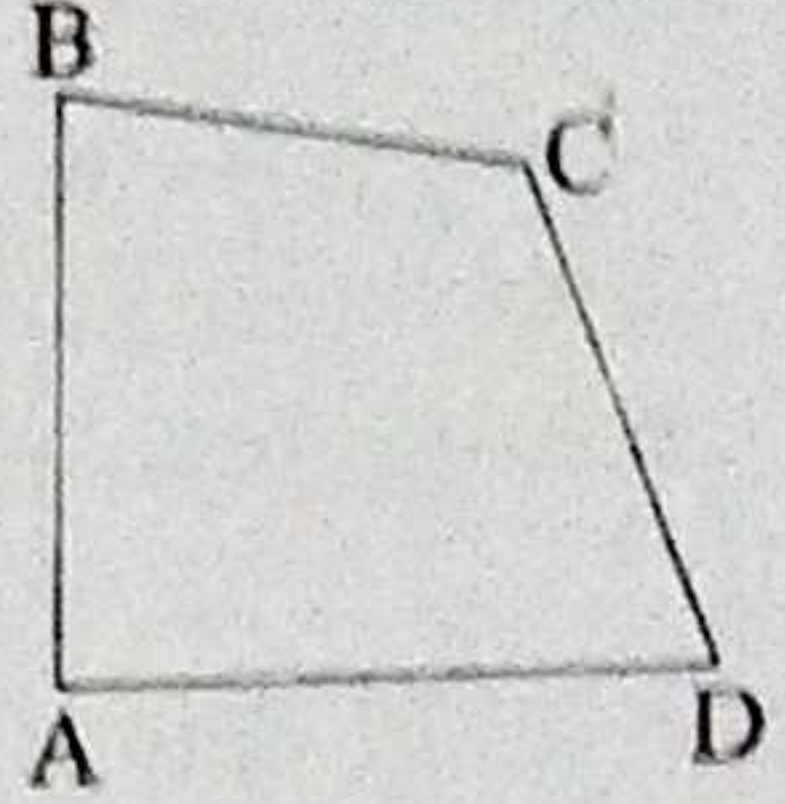
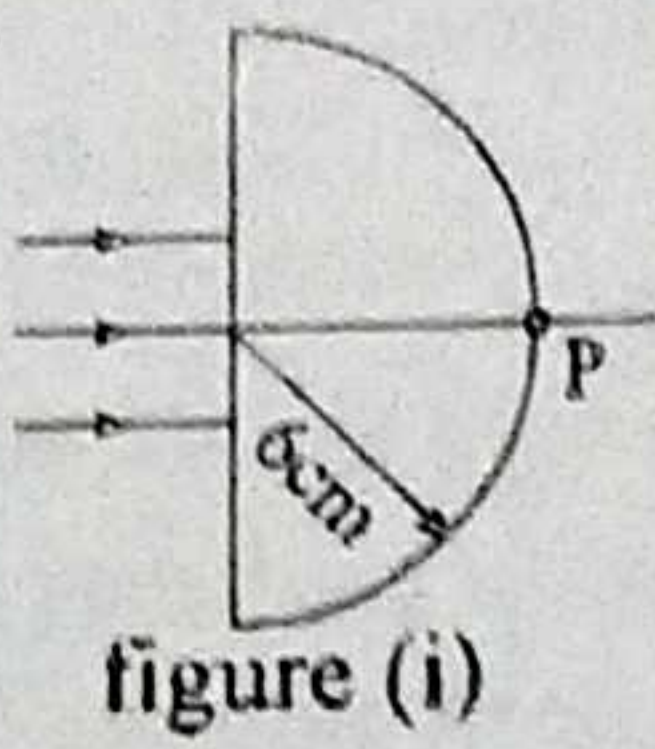
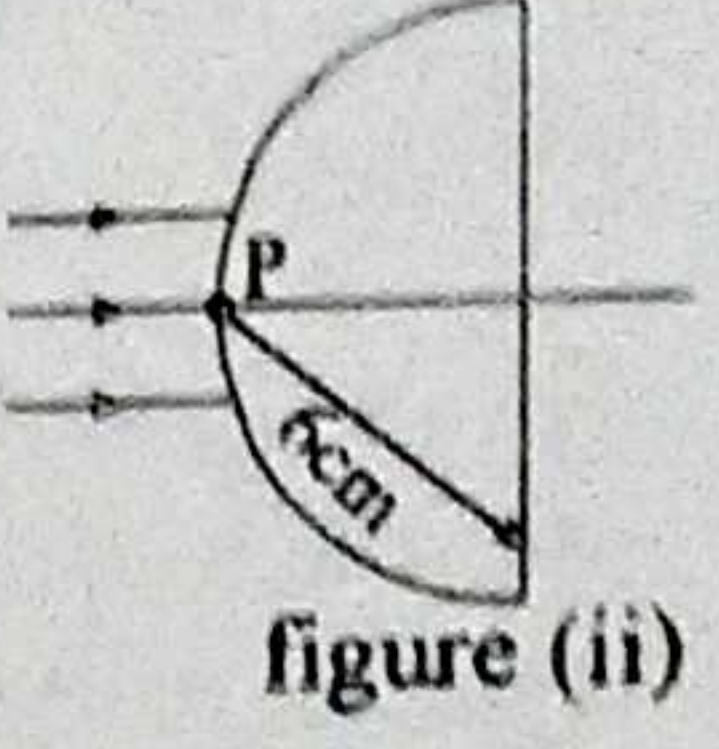
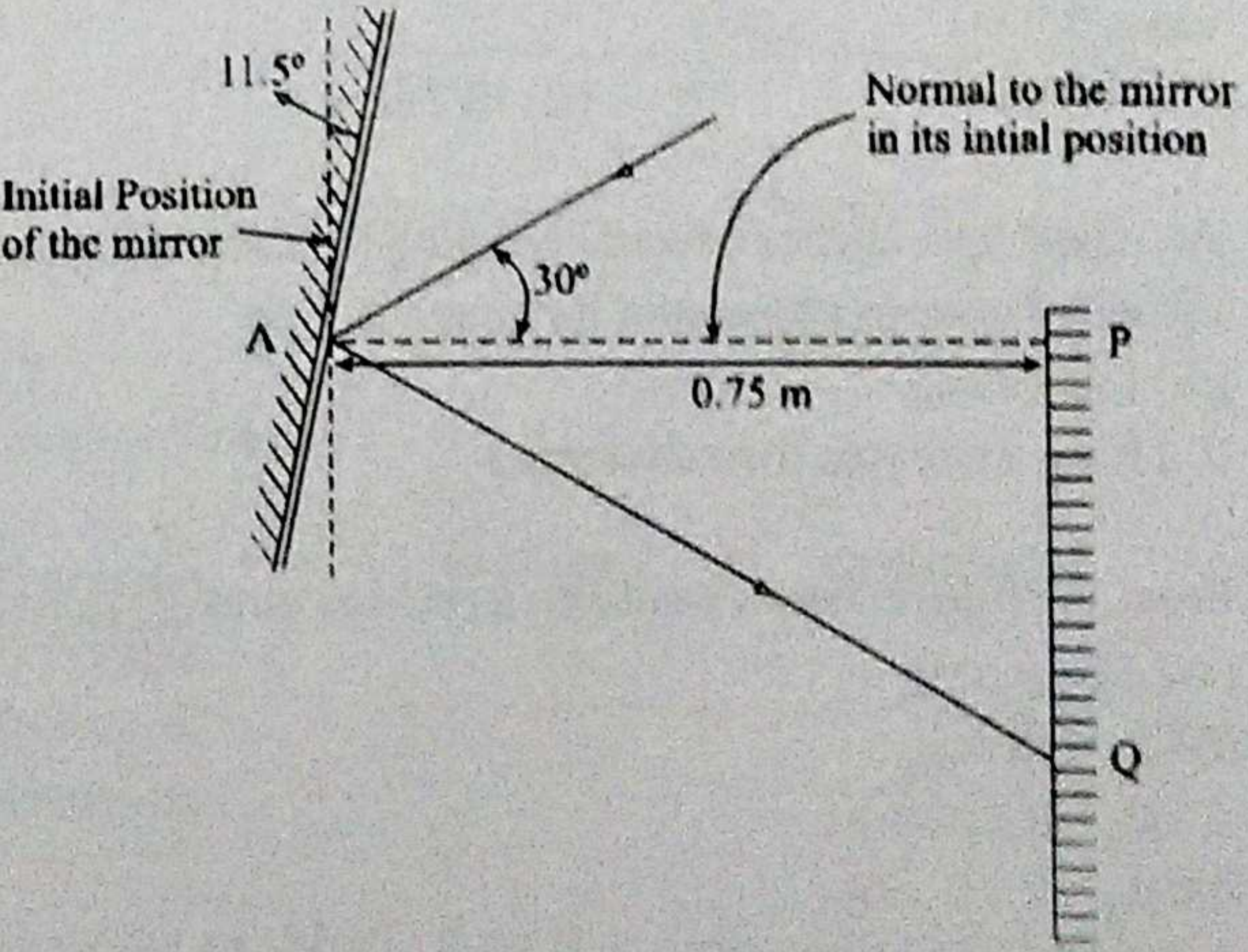
Geometrical Optics

Type of CPP - Objective & Subjective

Level - MODERATE

- In an experiment performed with a 60° prism where angle of minimum deviation for sodium light is 60° in air. The following experiment was done. When sodium light enters at one face at grazing incidence from a certain liquid, it emerges from the other face (in air) at 60° from the normal to edge of the prism. Are the observations correct?
- A ray of light is incident on a face of equilateral triangle at an incident angle 50° . At this angle minimum deviation occurs. This deviation is :
(A) 30 (B) 40 (C) 50 (D) None of these
- The image of the sun is obtained on a screen with a convex lens of diameter 3π cm and focal length 45 cm. How many times will the intensity produced by sun's image be greater than that coming directly from the sun if the angular diameter of the sun is 30 minutes ?
(A) 450 (B) 288 (C) 576 (D) None
- A ray of light in air is incident on face AB of an irregular block made of material with refractive index $\sqrt{2}$, as shown in figure. The face CD opposite to AB is a spherical surface of radius of curvature 0.4 m. From this face the refracted ray enters a medium of refractive index 1.514 and meets the axis PQ at point E . Determine the distance OE correct to two decimal places.
- The focal length of a lens of refractive index $3/2$ is 10 cm in air. The focal length of that lens in a medium of refractive index $7/5$ is :
(A) -70 cm (B) 10/7 cm (C) 70 cm (D) None of these
- An object approaches a fixed diverging lens with a constant velocity from infinity along the principal axis. The relative velocity between object and its image will be :
(A) Increasing (B) Decreasing
(C) First increases then decreases (D) First decreases and then increases
- From the figure shown establish a relation among μ_1, μ_2, μ_3 :
(A) $\mu_3 > \mu_2 > \mu_1$
(B) $\mu_3 < \mu_2 < \mu_1$
(C) $\mu_2 > \mu_3, \mu_3 = \mu_1$
(D) $\mu_2 > \mu_1, \mu_3 = \mu_2$
- The distance between a fixed object and the fixed screen is 100 cm. A lens produces an image on the screen when the lens is placed at either of the positions 40 cm apart. The power of the lens is nearly :
(A) 3 diopters (B) 5 diopters (C) 2 diopters (D) 9 diopters
- A converging lens of focal length 15 cm and a converging mirror of focal length 10 cm are placed 50 cm apart with common principal axis. A point source is placed in between the lens and the mirror at a distance of 40 cm from the lens. Find the locations of the two images formed.
- State true 'or' false : A biconvex lens may work as diverging lens.



11. **STATEMENT-1** : A ray is incident from outside on a glass sphere surrounded by air as shown. This ray may suffer total internal reflection at second interface.
STATEMENT-2 : For a ray going from denser to rarer medium, the ray may suffer total internal reflection.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
- 
12. The faces of prism $ABCD$ made of glass with a refractive Index n from dihedral angles $\angle A = 90^\circ$, $\angle B = 75^\circ$, $\angle C = 135^\circ$ & $\angle D = 60^\circ$ (The Abbe's prism). A beam of light falls on face AB & after total internal reflection from face BC escapes through face AD . Find the range of n and angle of incidence α of the beam onto face AB , if a beam that has passed through the prism in this manner is perpendicular to the incident beam.
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13. A parallel beam of light is incident normally on the flat surface of a hemisphere of radius 6 cm and refractive index 1.5, placed in air as shown in figure-(i). Assume paraxial ray approximation :
 (A) The rays are focussed at 12 cm from the point P to the right, in the situation as shown in figure-(i)
 (B) The rays are focussed at 16 cm from the point P to the right, in the situation as shown in figure-(i)
 (C) If the rays are incident at the curved surface (figure-(ii)) then these are focused at distance 18 cm from point P to the right.
 (D) If the rays are incident at the curved surface (figure-(ii)) then these are focused at distance 14 cm from point P to the right.
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14. A hemispherical portion of the surface of a solid glass sphere ($\mu = 1.5$) of radius r is silvered to make the inner side reflecting. An object is placed on the axis of the hemisphere at a distance $3r$ from the centre of the sphere. The light from the object is refracted at the unsilvered part, then reflected from the silvered part and again refracted at the unsilvered part. Locate the final image formed.
15. The view in the figure is from above a plane mirror suspended by a thread connected to the centre of the mirror at point A. A scale is located 0.75 m (the distance from point A to point P) to the right of the centre of the mirror. Initially, the plane of the mirror is parallel to the side of the scale; and the angle of incidence of a light ray which is directed at the centre of the mirror is 30° . A small torque applied to the thread causes the mirror to turn 11.5° away from its initial position. The reflected ray then intersects the scale at point Q. The distance from point P to point Q on the scale is :
 (A) 1.00 m
 (B) 0.56 m
 (C) 1.02 m
 (D) 0.86 m
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16. State true 'or' false : A particle moves towards the pole of a converging lens. The image moves away from lens wherever is the position of the object. (Except at focus)

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ANSWER KEY

1. [No] 2. (B) 3. (C) 4. [OE = 6.06 m] 5. (C) 6. (B) 7. (C) 8. (B)
 9. [One at 15 cm and the other at 24 cm from the lens away from the mirror] 10. [True] 11. (D)
 12. [$r + \theta = 75^\circ$; $\theta = 45^\circ$ (geometry), $\Rightarrow c < 45^\circ \Rightarrow 2 > n > \sqrt{2}$, $45^\circ < \alpha < 90^\circ$ (snell's law)] 13. (A)(D)
 14. [At the pole of reflecting surface of the sphere] 15. (A) 16. [False]